

TEMPLATE

KEY PROJECT INFORMATION & VPA DESIGN DOCUMENT (PDD)

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VERSION **v. 2.1**

RELATED SUPPORT Programme of Activity requirements and procedures

This document contains the following Sections

KEY PROJECT INFORMATION

SECTION A - DESCRIPTION OF PROJECT

SECTION B – APPLICATION OF APPROVED GOLD STANDARD METHODOLOGY (IES)
AND/OR DEMONSTRATION OF SDG CONTRIBUTIONS

SECTION C - DURATION AND CREDITING PERIOD

SECTION D – SUMMARY OF SAFEGUARDING PRINCIPLES AND GENDER SENSITIVE
ASSESSMENT

SECTION E - SUMMARY OF LOCAL STAKEHOLDER CONSULTATION

SECTION F - ELIGIBILITY AND INCLUSION CRITERIA FOR VPAS INCLUSION

Appendix 1 - Safeguarding Principles Assessment (mandatory)

Appendix 2- Contact information of VPA Implementer (mandatory)

Appendix 3- LUF Additional Information

Appendix 4-Summary of Approved Design Changes (VPA specific)

KEY PROJECT INFORMATION

Type of VPA	☒ Real case VPA (An actual activity implemented/proposed to be implemented under a PoA that sets out a specific regulatory and design framework to be followed by similar regular VPAs. It is distinguished with other real case VPAs based on aspects like; technology/measure types, host country, end user type etc) ☐ Regular VPA (An activity involving single measure or a set of interrelated measures implemented under a PoA that follow the framework/requirements set out by an associated real case VPA and PoA)
Scale of VPA Note that a VPA can be of one scale. Please select applicable scale accordingly.	☐ Microscale ☒ Small scale ☐ Large scale <i>Refer to corresponding Activity Requirements for definition of the scale of PoA. A regular VPA shall comply with the scale requirements as defined at corresponding real case VPA.</i>
Title of corresponding real case VPA (if applicable)	NA
GS ID of real case VPA (if applicable)	NA
GS ID of VPA	GS 11211
Title of VPA	UpEnergy – Social and Climate Impact Programme-Cooking Devices VPA-9
Time of First Submission Date	10/06/2021 and 9:00 pm
Date of Design Certification	26/04/2023
Version number of the VPA-DD	7
Completion date of version	18/08/2023
Coordinating/managing entity	UpEnergy Group
VPA Implementer (s)	Community Carbon

Project Participants and any communities involved	Community Carbon
Host Country (ies)	Mozambique
GS ID and Title of applicable Design Certified VPA	NA
GS ID and Title of applicable Performance Certified VPA	NA
Activity Requirements applied	☒ Community Services Activities (in general – off grid renewable energy, distributed technology, biogas, WASH) ☐ Renewable Energy Activities (in general – renewable energy projects connected to national or regional grids or industrial facilities) ☐ Land Use and Forestry Activities/Risks & Capacities ☐ N/A (projects that do not fall into either of the above, for example Shipping) Refer to the activity requirements for specific criteria
Other Requirements applied	☒ Programme of Activity requirements and procedures (PoA Requirements mandatory) Use this space to list requirements applicable to certain types of projects. E.g. microscale
Methodology (ies) applied and version number	Technologies and Practices to Displace Decentralized Thermal Energy Consumption (TPDDTEC), version 3.1
Product Requirements applied	☒ GHG Emissions Reductions & Sequestration (required to issue VERs/PERs and label CERs) ☐ Renewable Energy Label (required to label Renewable Energy Certificates) ☐ N/A (This is the rare case when a project chooses to issue neither emission reductions/labels or renewable energy labels)
VPA Cycle:	☐ Regular (Stakeholder Consultation (first round) has been conducted before the Start Date of the VPA) ☒ Retroactive (Stakeholder Consultation (first round) is conducted after the Start Date of the VPA) Retroactive VPAs can claim GS VERs/CERs for a max 2 years prior to the date of design certification.

Table 1 – Estimated Sustainable Development Contributions

Sustainable Development Goals Targeted	SDG Impact (defined in B.6)	Estimated Annual Average	Units or Products
13 Climate Action (mandatory)	Emission Reductions	23,368	tCO ₂ (eq)
1 (No poverty)	Percentage of users reporting money savings due to reduction in purchased fuel consumption in project	100	%
3 (Good Health and Well being)	Percentage of users reporting reduction in smoke/PM after shifting to ICS in project	100	%
5 (Gender Equality)	Percentage of users reporting time saving due to reduction in collected fuel consumption/cooking time	95	%
7 (Affordable and Clean Energy)	Number of households with operational stoves	14,780	-
8 (Decent Work and Economic Growth)	Total number of jobs created	40	-

12 (Responsible Consumption and Production)	Average percentage fuel savings reported by users in the project	55	%
15 (Life on Land)	Fuelwood equivalent savings reported by users in the project	113.81	tonnes-equivalent fuelwood/day

SECTION A. DESCRIPTION OF PROJECT

A.1. Purpose and general description of project

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The purpose of the VPA is to support clean cooking interventions such as electric cooktop/cooker and energy-efficient biomass improved cookstoves (ICS that moves end-users up the energy ladder and reduce greenhouse gas (GHG) emissions from the burning of non-renewable woody biomass and/or charcoal for cooking in Mozambique.

The VPA is implemented by Community Carbon & UpEnergy Group is the Coordinating and Managing Entity (CME) of the PoA. UpEnergy will implement the programme in partnership with local partners and would ensure the last-mile distribution/installation of the Improved Cookstoves to the beneficiaries.

More than 2.5 billion people in world lack access to clean cooking facilities, relying instead on solid biomass, kerosene or coal as their primary cooking fuel¹. According to statistics household air pollution, mostly from cooking smoke, is linked to around 2.5 million premature deaths annually worldwide.

The lack of access to clean cooking remains very acute in sub-Saharan Africa with only 17% population having clean cooking access. The vast majority of people in sub-

¹ IEA (2020), SDG7: Data and Projections, IEA, Paris <https://www.iea.org/reports/sdg7-data-and-projections>

Saharan Africa thus rely on gathering biomass for cooking, in particular in rural areas, which dramatically damages health and impairs productivity improvements. Almost 490 000 premature deaths² per year are related to household air pollution from the lack of access to clean cooking facilities, with women and children the worst affected. Forest degradation, sometimes leading to deforestation, is another serious consequence of the unsustainable harvesting of fuelwood. Clean cooking solutions such as Electric cooktop/cooker, avoids any use of non-renewable biomass/charcoal and thereby reducing GHG emissions. Improved Cookstoves (ICS) improve heat transfer efficiency as compared to the baseline conventional stoves, thereby reducing both the amount of fuel used by ICS and GHG emissions from the burning of non-renewable woody biomass and/or charcoal.

CME approves VPA-9 (GS-ID: 11211) to be included under the PoA 10963.

A.1.1. Eligibility of the VPA under approved PoA

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No.	Eligibility Criterion	Description / Required condition	Description of the VPA in relation to the criteria, Means of Verification and Supporting evidence for inclusion
1	Location of the VPA	The VPA is located within the geographical boundary of one of the countries included in the PoA.	Improved Cookstoves (ICS) ³ will be distributed to residential users within the geographic boundary of Mozambique which is within the boundary of PoA.
2	Type of project	Projects will involve the distribution of energy-efficient cooking technologies and safe drinking water systems	VPA includes distribution of ICS to residential users. Detailed information on technology is given in Section A.3 of VPA-DD.

² IEA (2020), SDG7: Data and Projections, IEA, Paris <https://www.iea.org/reports/sdg7-data-and-projections>

³ For simplicity, the remaining section of the document will mention Improved cookstove (ICS) and it is assumed it includes all the technologies such as Electric cooktop/cooker, and energy-efficient improved wood/charcoal stoves.

(HWT and IWT technologies) to residential users.

3	No Double counting of Clean Energy identification system for Products (CEPs) and VPAs the CEP installed is within this PoA and across applied. This shall other PoAs	A unique numbering or counting of stoves within the PoA and ensure that stoves can be identified as belonging to this PoA and not to a PoA managed by any other CME.	Each ICS distributed under the PoA will have the UpE logo or Brand name (Smart Home or similar) and unique serial number on the product. The unique serial number along with the customer details (name, address, is also stored in the sales database along with the unique serial number assigned to each product and the VPA assigned to it.
			At the point of sale, sales receipts are signed by the customers having a declaration that they are not part of any other programme. Customers are made aware about this clause before they sign the receipts. During the project implementation period, end users are also asked if they part of any other programme (ICS or WPS or Borehole projects); during the project implementation period, if it has been found that any of the end user is part of other project activity, that particular end user will not be considered in the ER calculation and will not be part of any of the project activity's ex-post surveys.

4	No Double counting of VPA The VPA is exclusively bound to the PoA. The VPA shall not be proposed as an individual Gold Standard or CDM project and/or as a part of any other CDM PoA and/or any other mechanism to avail climate change mitigation benefits. A statement shall be included in the VPA-DD that the specific VPA will not be part of another single Gold Standard or CDM project activity or VPA under another PoA and confirmed by the Partner Organization (PO) implementing the VPA.	The VPA is not part of another project/programme of activities under any carbon standards. The carbon standard registries including UNFCCC, GS and VERRA have been checked and it is confirmed that the VPA has not been registered as an individual Gold Standard or CDM project and/or as a part of any other CDM PoA and/or any other mechanism to avail climate change mitigation benefits.
5	Awareness and agreement of those operating a VPA on PoA subscription Contractual provisions to ensure that those operating the VPA are aware and have agreed that their activity is being subscribed to the PoA. In the case that the CME is not responsible for implementing the VPA, the organization responsible for VPA implementation has signed a contractual agreement with the CME to participate in the PoA. This agreement: --Defines the ownership of the carbon emission reduction rights --Covers the distribution and monitoring related	Community Carbon will be implementing the project

		responsibilities of the parties involved --Confirms that the CEPs to be distributed under the VPA have not and will not be distributed under any other carbon project (CDM project, PoA or Gold Standard project) --Cedes the rights to the carbon credits generated from VPAs under the PoA to the CME.	
6	Non-diversion of ODA in case of public funding	The CME and the VPA operator (in case of being different from the CME) shall confirm that there is no public funding or in the case of public funding, the annex I party will confirm that funding is not a diversion of Official Development Assistance.	It is confirmed that there is no diversion of ODA. An ODA declaration confirming the same has been submitted.
7	VPA Crediting Period	VPA crediting period not to exceed the PoA end date and the starting date of the crediting period of a VPA shall be on or after: (i) the VPA crediting period start date is either the VPA Start Date or two years prior to the date of Design Certification whichever is later.	The VPA's crediting period commenced on 05/07/2022. The VPA will have a crediting period of 5 years which can be renewed twice, i.e. in total a maximum period of 15 years. The VPA crediting period will not exceed the end date of the registered PoA.
8	Approval of VPA by CME	CME approves each VPA to be included into its registered PoA.	CME approves to include this VPA-DD into its registered PoA. CME declaration on this is shared.
9	Methodology Requirement	Each ICS VPA will comply with the applicability criteria of	The VPA complies with all applicability criteria of

	the applied methodology (TPDDTEC, version 03.1) and Safe drinking water VPA will comply with the applicability criteria of methodology emission reduction from safe drinking water supply-version 1.0	TPDDTEC, version 03.1. A detailed justification is provided in Section B.2 of VPA-DD.
10 Financial Additionality	Option-1: Projects located in LDC, LLDC, SIDS according to 4.1.9 (b) of 'Community Services Activity Requirements'. Option-2: All VPAs to be included under the PoA will be in compliance with item 1.1.3 of Annex B – positive list mentioned in the 'Community Services Activity Requirements', Version 1.2. All VPAs will be solely composed of isolated units (CEPs) where the users of the technology/ measure are households or communities and where each unit results in $\leq 600 \text{ MWh}/1.8\text{GWhth}$ of thermal energy savings per year or $\leq 600 \text{ tonnes}$ emission reduction per year	The VPA-DD is implemented in Mozambique which is LDC/LLDC, and according to 4.1.9(b) of the 'Community Services Activity Requirements' deemed additional.
11 Scale of the VPA	VPAs included under the PoA can be either small or large scale. In case of large scale VPAs, CME shall confirm that there is no	VPA-DD is a small scale VPA and adhering to the threshold of 180 GWhth for ICS

	suppressed demand claim. In case of small-scale threshold of 60,000 tCO₂/crediting year for safe water VPAs and 180 GWhth for ICS VPAs will be adhered.	
12 Project Technology	Specification of technology or measures, such as the level and type of service, as well as performance specification based on, intra alia, testing/certification	Project technology along with technical specifications is outlined in section A.3
13 Sampling	Sampling approaches are set out in each VPA and will follow the TPDDTEC v3.1 and emission reduction from safe drinking water supply-version 1.0 methodology. For safe water cross-VPA sampling a maximum of 10 VPAs can be grouped.	Sampling method is described in VPA-DD as per applied methodology requirements.
14 Multi-country PoA	In accordance with 17.1.2 “Programme of Activity Requirements”, an exception is approved for the development multi-country PoA. The VPA-DD developed should be part of this approved deviation request (Deviation reference- “POA-17.003”) batching. As per 17.1.4 of “PoA Requirements”, one VPA shall be submitted at the time of design certification and	The VPA-DD is developed for Mozambique which is one of the countries in the approved deviation request “POA-17.003”.

		subsequent VPAs for other countries shall be included at a later stage.	
15	SDG Outcomes	A minimum of two SDG outcome to be included in addition to SDG-13	All VPA-DDs will include a minimum of three SDG impact assessment including SDG-13. SDG targets and impact are described in section A.4 of the document

As per section 3.1.1 of GS4GG Principles & Requirements, compliance with relevant Eligibility criteria is demonstrated below:

Eligibility Criteria Category	Eligibility criterion - Required condition	Justification
1. Types of Project	Eligible projects shall include physical action/implementation on the ground. Pre-identified eligible project types are identified in the Eligibility Principles and Requirements section.	Project is already one of the pre identified types as per section 3.1.1 (b) and automatically eligible for Gold Standard Certification as per section 4.1.3 of GS4GG Principles & Requirements.
2. Location of Project	Projects may be located in any part of the world.	Location of the PoA is the Republic of Mozambique.
3. Project Area, Project Boundary and Scale	The Project Area and Project Boundary shall be defined. Projects may be developed at any scale although certain rules, requirements and limitations may apply under specific Activity Requirements, Impact Quantification Methodologies and Products Requirements. In order to avoid double counting the Project shall not be included in any other voluntary or compliance standards programme unless approved by Gold Standard (for example through dual certification). Also, if the Project Area overlaps with that of another Gold Standard or other voluntary or compliance standard programme of a similar nature, the Project shall demonstrate that there is no double counting of impacts at	The boundary for the PoA in terms of a geographical area is defined as the territorial boundary of Mozambique. All voluntary programme activities (VPAs) associated with this PoA will be implemented within the geographical boundary of the PoA. To avoid inclusion of any stove which is a part of another registered carbon project/ programme, all ICS under this programme shall have a unique ID number / Tag number, either inscribed on the stove or retained by the buyer, to uniquely identify the ICS avoiding any double counting and trace its user, later during monitoring and verification.

Eligibility Criteria Category	Eligibility criterion - Required condition	Justification
	design and performance certification (for example use of similar technology or practices through which the potential arises for double counting or misestimation of impacts amongst projects)	
4. Host Country Requirements	Projects shall be in compliance with applicable Host Country's legal, environmental, ecological and social regulations.	The PoA complies with Mozambique legal, environmental, and ecological and social regulations.
5. Contact Details	As part of the Project Documentation the Project Developer shall provide (i) name and (ii) contact details of all Project Participants; AND in case of an organisation (iii) the legal registration details and (iv) documentation by the governing jurisdiction that proves that the entity is in good standing (defined as being a legal or other appropriate entity registered in or allowed to operate within the required jurisdiction and with no evidence of insolvency or legal/criminal notices placed against it or any of its Directors). Gold Standard retains the right (at its own discretion) to refuse use of the Standard where reputational concerns are highlighted.	Name and Contact details of Project Participants is given in the Appendix 2. Legal registration details: Community Carbon is duly incorporated under the provisions of the Companies Act 2001 as Category 1 Global Business Company by Republic of Mauritius on 03/06/2022. Company Registration No: 188181. Relevant document is submitted
6. Legal Ownership	Full and uncontested legal ownership of any Products that are generated under Gold Standard Certification, (for example carbon credits) shall be demonstrated. Where such ownership is transferred from project beneficiaries this must be demonstrated transparently and with full, prior and informed consent (FPIC). Note that for certain Project types there is a requirement for full and	Criteria for transfer of carbon credit ownership: For regular cycle VPA, this shall be ensured through relevant provisions for example disclaimer on warranty/information cards, stove packaging, customer agreements / sales receipts / consent form or may be collected via monitoring app etc. or collecting stakeholder feedback on this issue

Eligibility Criteria Category	Eligibility criterion - Required condition	Justification
	uncontested legal land title/tenure to be demonstrated. These are contained within specific Activity or Product Requirements. All projects shall immediately report to Gold Standard any land title/tenure disputes arising.	during local stakeholder consultation (LSC) For retroactive VPA, this shall be ensured through relevant provisions for example disclaimer on warranty cards, stove packaging, customer agreements / sales receipts/ consent form or may be collected via monitoring app, etc. or stakeholder feedback collected during Stakeholder Feedback Round (SFR).
7. Other Rights	As well as legal title and ownership, the Project Developer shall also demonstrate where required uncontested legal rights and/or permissions concerning changes in use of other resources required to service the Project (for example, access rights, water rights etc.). Any known disputes or contested rights must be declared immediately to Gold Standard by the Project Developer and resolved prior to further project implementation in affected areas.	Not applicable
8. Official Development Assistance (ODA) Declaration	All Project Developers applying for project activities located in a country named by the OECD Development Assistance Committee's ODA recipient list and seeking Gold Standard Certification for carbon credits shall declare the Official Development Assistance (ODA) support. The Project Developer shall follow the GHG Emissions Reduction & Sequestration Product Requirements and submit the declaration at the time of Design Certification.	No ODA is involved in the PoA. A declaration is being submitted.

Eligibility under Gold Standard Community Services Activity (CSA) Requirements

As per section 3 of GS4GG Community Services Activity (CSA) Requirements, Eligibility criteria is defined below:

Eligibility Criteria Category	Eligibility criterion - Required condition	Justification
1. Eligible Project Types	All CSA Projects shall lead to climate change mitigation and/or adaptation by providing or improving access to services/resources at the household level. Eligible services include electricity and energy, water and sanitation, waste management, housing, etc.	The goal of the proposed PoA is to distribute ICS (improving access to services) in residential within the national borders of the Mozambique
2. GENERAL ELIGIBILITY CRITERIA - Type of project	(b) End-use energy efficiency: Project activities that reduce energy requirements as compared to baseline scenario without affecting the level and quality of services or products, where the end-user of the products and services are clearly identified and when the physical intervention is required at the user end. For example, efficient cooking, heating, lighting, etc.	The PoA involves distribution of energy efficient ICS to residential in Mozambique
3. GENERAL ELIGIBILITY CRITERIA – Project Area, Boundary and scale	Project Area and Boundary shall be defined in line with the applicable Impact Quantification Methodologies and Product Requirements.	The project area is point location of ICS beneficiaries in the host country of the VPA. The project boundary will be limited to the geographical boundary of the host country
4. GENERAL ELIGIBILITY CRITERIA – Legal Ownership	(a) Projects involving the distribution of a large number of devices for services such as heating, cooking, lighting, electricity generation, water treatment technology	The CEP owners will be transferring their rights on ownership of carbon credits to CME via the end user agreement /consent form via monitoring app etc

Eligibility Criteria Category	Eligibility criterion - Required condition	Justification
	such as water filter, etc. shall provide a clear description of the ownership of the Products that are generated under Gold Standard Certification all along the investment chain. In line with the FPIC requirement, the proofs that end-users are aware of and willing to give up their rights on Products shall be provided. (b) The transfer of Product ownership shall be discussed during local stakeholder consultations for projects.	The same will be discussed during stakeholder consultations.

A.1.2. Legal ownership of products generated by the VPA and legal rights to alter use of resources required to service the project

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Community Carbon has the legal ownership of the Verified Emission Reductions (VERs) that are generated through the Gold Standard Certification. The carbon title for the product is signed off by end user directly to Community Carbon waiving any claim or rights on carbon credits generated under the VPA.

A.2. Location of VPA

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Host Party: Mozambique

Region/State/Province: All across Mozambique

City/Town/Community: All across Mozambique

The geographical location of Mozambique is depicted below. Mozambique officially the Republic of Mozambique is a country located in Southeastern Africa bordered by the Indian Ocean to the east, Tanzania to the north, Malawi and Zambia to the northwest, Zimbabwe to the west, and Eswatini (Swaziland) and South Africa to the

southwest. with latitude $18^{\circ} 5' 14.877''$ S and longitude $34^{\circ} 45' 0.7092''$ E . The geographical map of Mozambique is shown in below figure A.1.



Figure 1 - The physical/geographical boundary of the SSC-PoA: Mozambique

A.3. Technologies and/or measures

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The VPA will include following distribution of Improved cookstoves to both residential and Institutional users. The programme will replace conventional biomass/charcoal stoves, with high efficiency efficient stoves (electric / biomass). The baseline stoves are “Three stone / traditional firewood / traditional charcoal stoves”. Improved cookstoves are more efficient than conventional biomass stoves as they reduce heat loss and improve heat transfer and/or combustion efficiency.

Sustainable Development Goals Targeted	How the project contributes to the identified SDG
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13 Climate Action (mandatory)	The emissions from the project stove are less than the baseline stove therefore reducing the GHG emitted.
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1 No Poverty	Projects leads to money saving from less fuel consumption in the cleaner technology
3 Good Health and Wellbeing	The project results in lesser emissions of CO/PM2.5 therefore improving the health of the end-users
5 Gender Equality	Project alleviates the strain on women by reducing the time spent on cooking and fuel collection
7 Affordable and Clean Energy	Project provides access to affordable and cleaner technology for cooking
8 Decent Work and Economic Growth	The project involves distribution of cleaner technology which in turn means number people will be employed to make the programme a success.
12 Responsible Consumption and Production	Project results in reduction in fuel consumption as compared to baseline
15 Life on Land	Project results in reduction in fuel consumption as compared to baseline which in turns reduce deforestation

The models and products to be distributed under the VPA are as follows:

1. Electric Models

The below examples are just indicative, the actual model specification can be different than below and shown in figure below. New electric cookstove models can be introduced at any time during project implementation based on end-user response.

- a. Model: SmartHome Electric pressure cooker
 - Capacity: 6 liters
 - Power: 1000-1190 watt
 - Voltage: 220-240V 50/60 Hz
 - Size: 38 x 34 x 33 cm
 - Weight: 5.9 kg
 - Time and Power savings: 70% faster than conventional cooking
 - Inner Pot: Aluminum non-stick pot

- Average Lifespan: 5 years



Figure 2: SmartHome Electric pressure cooker

2. Improved Cookstoves (biomass)

The improved cookstoves reduce fuel consumption through use of a ceramic liner that increases combustion efficiency and retains heat. The stoves consist of a metal frame (called cladding) with interior ceramic liner for thermal insulation. The ICS has a provision that allows ash collection at the base. A thin layer of cement is placed between the cladding and the liner to bind the two. During use, a single pot rests at the top the stove. The fixed/inbuilt types are finished with either selected bricks, tiles or stainless steel. The movable types are of metallic stainless-steel cladding. The insulations include of pumice, mica, vermiculite and are bonded with ordinary cement. The smoke extracts through the chimney pipes which are metallic. The fitting stainless steel saucepans are

of the thickness 3mm bottom and 2mm sides. The average lifespan of the improved cookstoves with residential use is 5 years⁴.

a. Examples of residential improved cookstoves are given below



Figure 3a: Residential Cookstoves



The technical specification of few models is shown in below table. Other models will be distributed in the PoA during the course of its crediting period.

Product Model	Dimension (In cm)	Avg. unit weight (In kg)	Thermal Efficiency (In %)	Estimated life (in years)	Fuel Type
Smart Home Pro	(Ø 28 x 26 H)	12.00	37.90	5	Charcoal
Community Cookstoves	(Ø 28 x 26 H)	12.00	38.70	5	

⁴ The lifetime of the ICS depends on usage rates and maintenance and may go beyond 5 years.

RH Stove	(Ø 26.5 x 22 H)	12.00	36.30	5	
TG Stove ⁵	(Ø 27 x 23.5 H)	12.50	35.90	5	
SpendSmart Stove (SST)	(Ø 23.9 x 24.4 H)	6.87	36.30	5	
EE Stove	(Ø 23.93 x 24.43 H)	8.40	33.00	5	

Other models for all cooking technologies for residential may also be distributed in this VPA over time.

A.4. Scale of the VPA

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The VPA is a small-scale project activity. The applied TPDDTEC methodology allows small scale project as the energy saving in the VPA is less than threshold limit of 180 GW_{th}/year.

A.5. Funding sources of VPA

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There is no public funding for the PoA. No ODA funding will be used, as confirmed by signed ODA Declarations submitted to GS.

SECTION B. APPLICATION OF APPROVED GOLD STANDARD METHODOLOGY (IES) AND/OR DEMONSTRATION OF SDG CONTRIBUTIONS

B.1. Reference of approved methodology (ies)

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⁵ The efficiency value for TG stove is considered as the median from which +-5% is calculated

- Technologies and Practices to Displace Decentralized Thermal Energy Consumption, version 3.1.
- Requirements and Guidelines for carrying out usage surveys for projects implementing improved cooking devices version 2 .

Tool:-

- CDM Methodological tool 5: Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation, version 03.0.
- CDM Methodological tool 30: Calculation of the fraction of non-renewable biomass, Version 03.0.

B.2. Applicability of methodology (ies)

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The table below lists each of the methodology applicability requirements and how the VPA complies with each of the criteria:

S.No	Methodology applicability requirement	Justification
1	This methodology is applicable to programmes or activities introducing technologies and/or practices that reduce or displace greenhouse gas (GHG) emissions from the thermal energy consumption of households and non-domestic premises	This VPA includes distribution of efficient improved cookstoves (ICS) reducing greenhouse gas (GHG) emissions from thermal energy consumption due to burning of non-renewable woody biomass and/or charcoal for cooking in Mozambique for residential users.
2.	The project activity is implemented by a project proponent and can include additional project participants. The individual households do not act as project participants.	This VPA is implemented by Community Carbon. UpEnergy is the CME of the PoA. The individual households do not act as project participants.
3.	The project boundary needs to be clearly identified, and the technologies	The geographical project boundary of this VPA is defined as the country of

	counted in the project are not included in any other voluntary market or CDM project activity (i.e. no double counting takes place). In some cases there maybe another similar activity within the same target area. Project proponents must therefore have a survey mechanism in place together with appropriate mitigation measures so as to prevent any possibility of double counting.	Mozambique (also detailed in Section A.2 of this document). The carbon standard registries including UNFCCC, GS and VERRA have been checked and it is confirmed that the VPA has not been registered as an individual Gold Standard or CDM project and/or as a part of any other CDM PoA and/or any other mechanism to avail climate change mitigation benefits. Hence, it can be confirmed that double counting is being avoided. Each ICS distributed under the PoA will have the UpE logo and unique serial number on the product. The unique serial number along with the customer details (name, address, is also stored in the sales database along with the unique serial number assigned to each product and the VPA assigned to it.
4.	The technologies each have continuous useful energy outputs of less than 150kW per unit (defined as the total useful energy delivered from start to end of operation of a unit divided by time of operation). For technologies or practices that do not deliver thermal energy in the project scenario but only displace thermal energy supplied in the baseline scenario, the 150kW threshold applies to the displaced baseline technology.	The ICS distributed under this VPA have a capacity of less than the maximum 150kW per unit. The same has been demonstrated in the ER calculation sheet

5.	Using the baseline technology as a backup or auxiliary technology in parallel with the improved technology introduced by the project activity is permitted as long as a mechanism is put into place to encourage the removal of the old technology (e.g. discounted price for the improved technology) and the definitive discontinuity of its use. The project documentation must provide a clear description of the approach chosen and the monitoring plan must allow for a good understanding of the extent to which the baseline technology is still in use after the introduction of the improved technology.	As such, users will be encouraged to discontinue and remove the baseline stoves. However, if some of the users still continue to use the baseline stoves, the same shall be captured during the annual monitoring surveys and will be accounted for in the emission reduction calculations as part of the project Kitchen Performance Test.
6.	The project proponent must clearly communicate to all project participants the entity that is claiming ownership rights of and selling the emission reductions resulting from the project activity. For technology producers and the retailers of the improved technology or the renewable fuel in use, this must be communicated by contract or clear written assertions in the transaction paperwork. If the claimants are not the project technology end users, the end users will need to be informed and notified that they cannot claim for emission reductions from the project.	Community Carbon will ensure that all local partners (if any) and end users are aware of the fact that Community Carbon is claiming ownership rights of and selling the emission reductions resulting from the project activity. It will be communicated to end users at the time of installation/ distribution as part of the sales receipt.
7.	Project activities making use of a new biomass feedstock in the project	The project activity complies with relevant Gold Standard specific

	situation (e.g. shift from non-renewable to green charcoal, plant oil or renewable biomass briquettes) must comply with relevant Gold Standard specific requirements for biomass related project activities, as defined in the latest version of the Gold Standard rules. If the biomass feedstock is sourced from a dedicated plantation, the criteria must apply to both plantations established for the project activity AND existing plantations that were established in the context of other activities but will supply biomass feedstock.	requirements for biomass related project activities, as defined in the latest version of the Gold Standard rules.
8.	Adequate evidence is supplied to demonstrate that indoor air pollution (IAP) levels are not worsened compared to the baseline, and greenhouse gases emitted by the project fuel/stove combination are estimated with adequate precision. The project fuel/stove combination may include instances in which the project stove is a baseline stove.	The fuel used in both the project and baseline scenario is the same, as such there are no additional harmful gases released in the project scenario. By disseminating improved cookstoves, the project will ensure that households consume less firewood for cooking, which will lead to a reduction of indoor air pollution levels.
9.	Records of renewable fuel sales may not be used as sole parameters for emission reduction calculation, but may be used as data informing the equations in section 2.0 of the methodology. These records need to be correlated to data on distribution and results of field tests and surveys confirming (a) actual use of the renewable fuel and usage patterns (such as average fraction of non-renewable	Not applicable since this criterion is applicable only in case of introduction of a new biomass feedstock.

	fuels used in mixed combustion or seasonal variation of fuel types), (b) GHG emissions, (c) evidence of CO levels not deteriorating (d) any further factors effecting emission reductions significantly.	
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The table below lists “Global Goals Community Services Activity Requirements”, version 1.2 requirements and how the VPA complies with the criteria:

S.No	Community Services Activity Requirements	Justification
1	Para 3.1.1: All CSA Projects shall lead to climate change mitigation and/or adaptation by providing or improving access to services/resources at the household level. Eligible services include electricity and energy, water and sanitation, waste management, housing, etc.	This VPA includes distribution of efficient improved cookstoves (ICS) reducing greenhouse gas (GHG) emissions from thermal energy consumption due to burning of non-renewable woody biomass and/or charcoal for cooking in Mozambique for residential users. Therefore, the project falls under Type b (End-use energy efficiency) of Pre-identified CSA project types.
2	Para 3.1.2: Project Area and Boundary shall be defined in line with the applicable Impact Quantification Methodologies and Product Requirements	The geographical project boundary of this VPA is defined as the country of Mozambique (also detailed in Section A.2 of this document).
3	Para 3.1.3: Certain Impact Quantification methodologies allow projects to account Suppressed Demand scenario when establishing a baseline. In such cases, the application of Suppressed Demand baseline is	This VPA is a small scale project.

	limited to Small Scale and Microscale Projects.	
4	Para 3.1.4: Projects involving the distribution of a large number of devices for services such as heating, cooking, lighting, electricity generation, water treatment technology such as water filter, etc. shall provide a clear description of the ownership of the Products that are generated under Gold Standard Certification all along the investment chain. In line with the FPIC requirement, the proofs that end-users are aware of and willing to give up their rights on Products shall be provided. The transfer of Product ownership shall be discussed during local stakeholder consultations for projects.	Community Carbon will ensure that all local partners (if any) and end users are aware of the fact that Community Carbon is claiming ownership rights of and selling the emission reductions resulting from the project activity. It will be communicated to end users at the time of installation/ distribution and during local stakeholder consultations for projects. The carbon title for the product will be signed off by end user to Community Carbon waiving any claim or rights on carbon credits generated under the VPA.

B.3. VPA boundary

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The project boundary is all across Mozambique. The country is a landlocked country in southeastern Africa bordered by the Indian Ocean to the east, Tanzania to the north, Malawi and Zambia to the northwest, Zimbabwe to the west, and Eswatini (Swaziland) and South Africa to the southwest. with latitude 18° 5' 14.877" S and longitude 34° 45' 0.7092" E . Schematic representation of baseline and project cookstoves is given in figure B.1 and B.2 below.

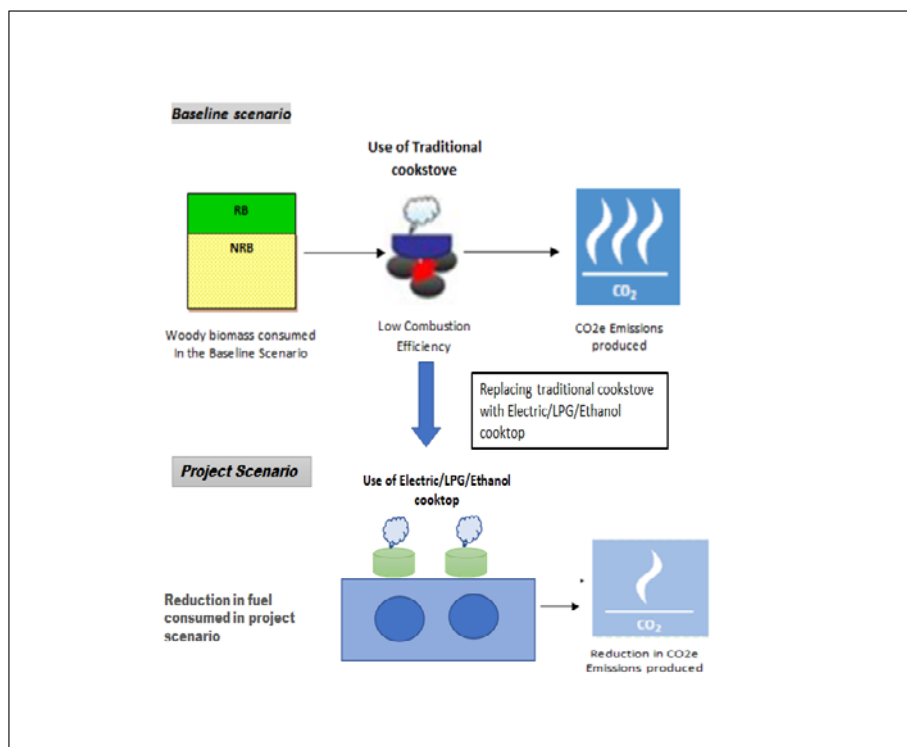


Figure B.1: Flow Diagram: baseline and project scenarios (Electric cooking)

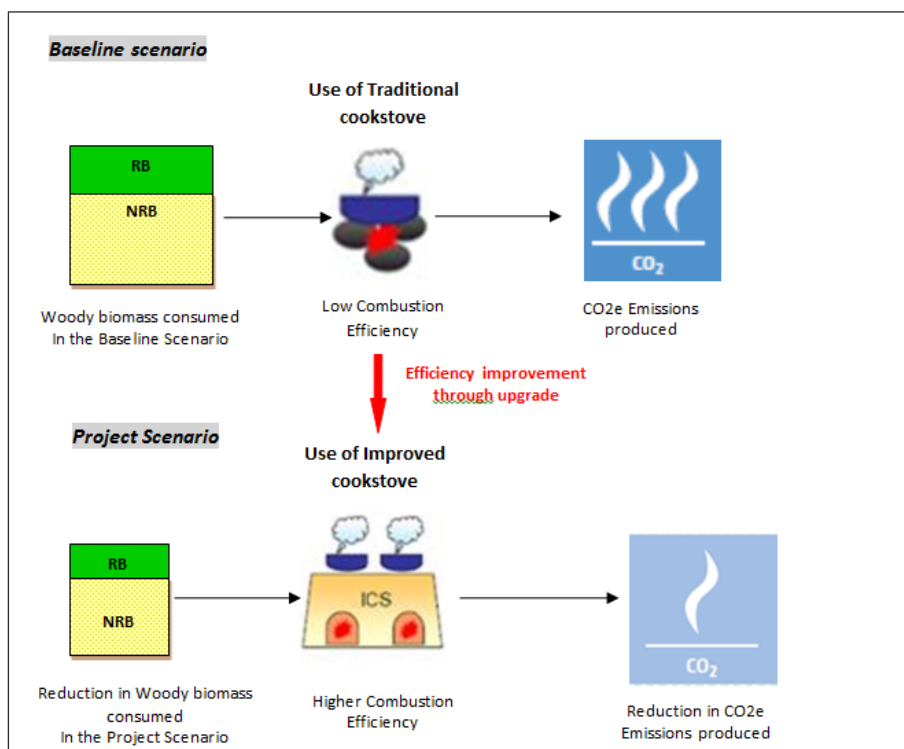


Figure B.2: Flow Diagram: baseline and project scenarios (ICS)

Source		GHGs	Include d?	Justification / Explanation
Baseline scenario	Combustion of non-renewable biomass for cooking	CO ₂	Yes	Main source of emission in the baseline
		CH ₄	Yes	Important source of emissions
		N ₂ O	Yes	Important source of emissions
Project scenario	Combustion of non-renewable biomass for cooking	CO ₂	Yes	Main source of emission in the project scenario
		CH ₄	Yes	Important source of emissions
		N ₂ O	Yes	Important source of emissions
	Consumption of electricity for cooking	CO ₂	Yes	Main source of emission in the project scenario
		CH ₄	No	Excluded
		N ₂ O	No	Excluded

B.4. Establishment and description of baseline scenario

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The baseline scenario as per the applied methodology TPDDTEC, version 3.1, is defined by the typical baseline fuel consumption patterns in a population that is targeted for adoption of the project technology.

Identification of baseline scenario included two methods – literature review and field/baseline survey. Literature and historical data review were done to identify the baseline cooking regime and fuel consumption pattern of the households.

Baseline Survey and Fuel Consumption

Baseline Study

The baseline scenario was established through the literature review. For literature study recognized journals/articles and reports of Demographic and Health Survey (DHS) where the surveys were also carried out in person with a very large sample size was used. The comprehensive questionnaire was designed by Demographic and Health Survey (DHS) and World Bank (Sustainable Energy for all (SE4All) to capture all the information of household like family members, age group, gender, address, contact

number, geographical coordinates, type of fuel & consumption and cookstove, sourcing of fuel long with time, health related issues, etc.

The baseline survey covered the following elements:

- Fuel Consumption Pattern.
- Available technology options used for cooking
- Classified the most promising biomass (wood, Charcoal, electricity, etc.) utilized during cooking.

The baseline survey followed a stratified two-stage methodology:

- The first stage involved selecting sample points (clusters) consisting of Enumeration area (EA). EAs were selected with a probability proportional to their size within each sampling stratum. Maximum clusters were selected.
- The second stage involved systematic sampling of households. A household listing operation was undertaken in all the selected clusters. During the listing, an average household were identified in each cluster, from which a fixed number of households were selected through an equal probability systematic selection process, to obtain a maximum sample size of households. Results from this sample are representative at the national, urban, and rural, and provincial levels.

All women aged 15-49 and men aged 15-59 who were either permanent residents of the selected households or visitors who stayed in the households the night before the survey were interviewed.

The survey in person interview was conducted in various households in Mozambique. The survey is designed to provide estimates at the national level, for urban and rural areas, and for the fuel consumption pattern. According to the baseline survey the average household size in Mozambique is 4.5 persons.⁶

⁶ <https://dhsprogram.com/pubs/pdf/AIS12/AIS12.pdf>

In Mozambique, 98% of the population from the study are using biomass for cooking out of which 75.1% of the population using fuelwood for cooking while 18.9% of the population using charcoal for cooking as the main cooking fuel. 3.7% have access to clean cooking fuel in rural and Urban. The penetration rate is 0.3% for the improved cookstove.

This VPA will replace conventional inefficient stoves, with higher efficiency ICS to residential users by leveraging resources provided by the PoA. Therefore, it is assumed that in the absence of the project activity, the baseline scenario would be the traditional cooking technologies with use of non-renewable biomass/charcoal for meeting similar thermal energy needs share.

In Mozambique, access to electricity remains a challenge. Three quarters (76%) of household's family members do not have access to electricity.⁷ The percentage varies depending on the area of residence, being the area, which has the highest percentage (68%) of households with access to electricity compared to rural areas (5%). Most families use solid fuel for cooking (97%) and half of the families (51%) cook outside the home⁸.

As per DHS Report Mozambique (2015), 5.6% children were with symptoms of acute respiratory infection (ARI) by using charcoal and 4.6% children affected ARI by using wood as fuel in households for cooking.⁹

Fuel gathering consumes most of the time of the household daily for women and children, limiting other productive activities (e.g., income generation) and taking children away from school. In less secure environments, women and children are at risk of injury and violence during fuel gathering.

⁷ <https://dhsprogram.com/pubs/pdf/AIS12/AIS12.pdf>

⁸ <https://dhsprogram.com/pubs/pdf/AIS12/AIS12.pdf>

⁹ <https://dhsprogram.com/pubs/pdf/AIS12/AIS12.pdf>

More than 99% of the target population for this VPA are using traditional biomass stoves with less thermal efficiency, emitting smoke and unhealthy environment inside the kitchen in both rural and urban segments.¹⁰

The comprehensive questionnaire was designed to capture all the information of household like family members, age group, gender, address, contact number, geographical coordinates, type of fuel & consumption and cookstove, sourcing of fuel long with time, health related issues, etc.

The household surveyed were not using improved cookstove using biomass as fuel. The penetration of ICS is very minimal in Mozambique. Clean and improved cookstoves have the opportunity to improve several parts of a population's health and well-being.

Baseline Fuel Consumption

A baseline Kitchen Performance test (KPT) has been made for identifying the average charcoal consume in the project area before the project activity. The survey was made in 2015 by GIZ (Deutsche Gesellschaft für Internationale Zusammenarbeit).

Result

As a result of the baseline KPT study it was found out that the mean charcoal consume by household was averagely 70 kg/HH/month.

Fuel Type	Quantity of fuel consumed, Kg/HH/month	Quantity of fuel consumed, MT/HH/day ¹¹	Quantity of eq. firewood consumed, MT/HH/day
Charcoal	70	0.00233	0.014 ¹²

¹⁰ <https://dhsprogram.com/pubs/pdf/AIS12/AIS12.pdf>

¹¹ BKPT will be conducted following paired sample requirements when the project implements for Residential users with similar baseline scenario. As per methodology, emission reduction will not be claimed until baseline/project scenario KPTs are conducted. This can be carried out post registration.

¹² Charcoal fuel consumption is multiplied by 6 as per the methodology which allows 1 quantity of charcoal equivalent to 6 quantity of fuel-wood

It can be seen from the above table that 0.014 MT of equivalent firewood consumed every day from each household which is 5.11 MT/HH/yr.

Baseline Non-renewable Biomass Assessment

The fraction of non-renewable biomass is determined based on the CDM Tool 30 – Calculation of the fraction of non-renewable biomass. The value of fractional non-renewability is 0.78.

This VPA will replace conventional inefficient stoves, with higher efficiency ICS to residential users by leveraging resources provided by the PoA. Therefore, it is assumed that in the absence of the project activity, the baseline scenario would be the traditional cooking technologies with use of non-renewable biomass/charcoal for meeting similar thermal energy needs share.

B.5. Demonstration of additionality

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Specify the methodology, activity requirement or product requirement that establishes deemed additionality for the proposed project (including the version number and the specific paragraph, if applicable).	Community Services Activity Requirements (Version 1.2), paragraph 4.1.9: “Projects that meet any of the following criteria are considered as deemed additional and therefore are not required to prove Financial Additionality at the time of Design Certification: (a) Positive list (Annex B) (b) Projects located in LDC, SIDS, LLDC (c) Micro-scale projects”
Describe how the proposed project meets the criteria for deemed additionality.	Project activities under the VPA are solely composed of isolated units where the users of the technology/measure are households and where each unit results in – ICS: <= 600 MWh of energy savings per year or WPS: <=600 tonnes of emission reductions per year. Therefore, 4.1.9, (a) as per the Community Services Activity Requirements is met.

B.5.1. Prior Consideration

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Not Applicable

B.5.2. Ongoing Financial Need

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Not applicable as this is only required at time of renewal of crediting period.

B.6. Sustainable Development Goals (SDG) outcomes

Relevant Target/Indicator for each of the three SDGs

Sustainable Development Goals Targeted	Most relevant SDG Target	SDG Impact Indicator (Proposed or SDG Indicator)
13 Climate Action (mandatory)	N/A	Emission Reductions
1 End poverty in all its forms everywhere	1.1 By 2030, ensure that all men and women, in particular the poor and the vulnerable, have equal rights to economic resources, as well as access to basic services, ownership and control over land and other forms of property, inheritance, natural resources, appropriate new technology and financial services, including microfinance	Percentage of households confirming money saving with clean energy products

3 Good Health and Well-being	3.9 By 2030, substantially reduce the number of deaths and illnesses from hazardous chemicals and air, water and soil pollution and contamination	Percentage of users reporting reduction in smoke/PM after shifting to ICS in project.
5: Gender Equality	5.4 Recognize and value unpaid care and domestic work through the provision of public services, infrastructure and social protection policies and the promotion of shared responsibility within the household and the family as nationally appropriate	Percentage of users reporting time saving due to reduction in collected fuel consumption/cooking time.
7: Affordable and Clean Energy	7.1 By 2030, ensure universal access to affordable, reliable and modern energy services	Number of households with operational stove
8: Decent Work and Economic Growth	8.3 Promote development-oriented policies that support productive activities, decent job creation, entrepreneurship, creativity and innovation, and encourage the formalization and	Total no of jobs created

	growth of micro-, small- and medium-sized enterprises, including through access to financial services	
12: Responsible Consumption and Production	12.2 By 2030, achieve the sustainable management and efficient use of natural resources	Average percentage fuel savings reported by users in the project.
15: Life on Land	15.2 By 2020, promote the implementation of sustainable management of all types of forests, halt deforestation, restore degraded forests and substantially increase afforestation and reforestation globally	Fuelwood equivalent savings reported by users in the project

B.6.1. Explanation of methodological choices/approaches for estimating the SDG Impact

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The VPA includes distribution of efficient Improved Cookstoves (ICS) with the aim to reduce greenhouse gas (GHG) emissions from the burning of non-renewable woody biomass and/or charcoal for cooking in Mozambique. The outcome of the SDG 13 (Climate Action) will be measured as reduced greenhouse gas emissions measured as tonnes of CO₂e applying the GS methodology TPDDTEC, version 3.1. The SDG 13 outcome will be certified as “Certified SDG 13 Impacts” allowing the generation of carbon credits (GS VERs).

When the baseline fuel and the project fuel are the same and the baseline emission factor and project emission factor are same, the overall GHG reductions achieved by the project activity in year y are calculated as follows:

$$ER_y = \sum_{b,p} (N_{p,y} * U_{p,y} * P_{p,b,y} * NCV_{b, fuel} * (f_{NRB,b, y} * EF_{fuel, CO2} + EF_{fuel, nonCO2})) - \sum LE_{p,y} \quad (1)$$

Where:

$\sum_{b,p}$	Sum over all relevant (baseline b/project p) couples
$N_{p,y}$	Cumulative number of project technology-days included in the project database for project scenario p against baseline scenario b in year y
$U_{p,y}$	Cumulative usage rate for technologies in project scenario p in year y, based on cumulative adoption rate and drop off rate revealed by usage surveys (fraction)
$P_{p,b,y}$	Specific fuel savings for an individual technology of project p against an individual technology of baseline b in year y, in tons/day, as derived from the statistical analysis of the data collected from the field tests
$NCV_{b,fuel}$	Net calorific value of the fuel that is substituted or reduced (IPCC default for wood fuel, 0.0156 TJ/ton)
$EF_{b,fuel,CO2}$	CO2 emission factor of the fuel that is substituted or reduced. 112 tCO2/TJ for Wood/Wood Waste, or the IPCC default value of other relevant fuel
$EF_{b,fuel,nonCO2}$	Non-CO2 emission factor of the fuel that is reduced 9.46
$LE_{p,y}$	Leakage for project scenario p in year y (tCO2e/yr)

When the baseline fuel and the project fuel are different and/or the emission factors are different, the overall GHG reductions achieved by the project activity in year y are calculated as follows:

$$ER_y = \sum_{b,p} N_{p,y} * U_{p,y} * (f_{NRB,b, y} * EF_{fuel, CO2} + EF_{fuel, nonCO2}) - \sum LE_{p,y} \quad (2)$$

Where:

$\sum_{b,p}$	Sum over all relevant (baseline b/project p) couples
$N_{p,y}$	Cumulative number of project technology-days included in the project database for project scenario p against baseline scenario b in year y
$U_{p,y}$	Cumulative usage rate for technologies in project scenario p in year y, based on cumulative adoption rate and drop off rate revealed by usage surveys (fraction)
$f_{NRB,b, y}$	Fraction of biomass used in year y for baseline scenario b that can be established as non-renewable biomass (drop this term from the equation when using a fossil fuel baseline scenario)

$EF_{b, fuel, CO2}$	CO2 emission factor of the fuel that is substituted or reduced. 112 tCO ₂ /TJ for Wood/Wood Waste, or the IPCC default value of other relevant fuel
$EF_{b, fuel, nonCO2}$	Non-CO2 emission factor of the fuel that is reduced
$LE_{p,y}$	Leakage for project scenario p in year y (tCO ₂ e/yr)

$$ER_y = \sum_{b,p} N_{p,y} * U_{p,y} * (f_{NRB,b,y} * ER_{b,p,y, CO2} + ER_{b,p,y, non-CO2}) - \sum LE_{p,y}$$

Where

Calculation for units including electric stoves:

The emission reduction achieved by the project activity shall be calculated as the difference between the baseline emission in the premises evaluated as per the baseline fuel consumption savings (accounting for any parallel use of baseline technology) and the project emissions evaluated as per the guidance provided in the CDM Methodological tool for Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation, version 3, as follows. The CDM tool is used to estimate the project emissions from electricity consumptions as per below equation. The baseline emissions are calculated using equation in TPDDTEC v3.1 as the baseline scenario for improved cookstove and electric stoves are the same. The leakage emission from project is zero as there is no net increase in electricity consumption other than project activity need which are already accounted in project emissions.

The monitoring of quantity of electricity consumed by project activity (electric cooking device) is determined by conducting 3-day consecutive kitchen performance test in which the energy meter will be fixed to the device to monitor the consumption of electricity for entire day. On the second day, record the final reading before start the cooking. This will be repeated for the three days. The fourth day the final reading is captured. The average reading of three days provided the consumption of electricity per day.

Project emissions due to electricity consumption from the grid is applicable for this project.

$$PE_y = \sum_d EG_{p,d,y} \times EF_{el,y} \times (1 + TDL_{j,y}) \quad (3)$$

$$PE_y = B_{ep,y} * ((f_{NRB,y} * EF_{p,fuel,CO2}) + EF_{p,fuel,nonCO2}) * NCV_{p,fuel}$$

Where:

Σ_d	Sum over all relevant devices
PE_y	Project emissions from electricity consumption in year y (t CO ₂ /yr)
$EG_{p,d,y}$	Quantity of electricity consumed by the project device d in year y (MWh/yr)
$EF_{el,y}$	Emission factor for electricity system in year y (t CO ₂ /MWh)
$TDL_{j,y}$	Average technical transmission and distribution losses for providing electricity to source j in year y
J	Sources of electricity consumption in the project
PE_y	Project emissions from the use of biomass in year y (t CO ₂ /yr)
$B_{ep,y}$	Quantity of fuel consumed by baseline technology in project scenario p during year y, in tons
$f_{NRB,y}$	Fraction of biomass used in year y that can be established as non-renewable biomass
$NCV_{p,fuel}$	Net calorific value of the fuel that is substituted or reduced (IPCC default for wood fuel, 0.0156 TJ/ton)
$EF_{p,fuel,CO2}$	CO ₂ emission factor of the fuel that is substituted or reduced. 112 tCO ₂ /TJ for Wood/Wood Waste, or the IPCC default value of other relevant fuel
$EF_{p,fuel,nonCO2}$	Non-CO ₂ emission factor of the fuel that is reduced

Calculations for Improved Cookstoves using woody biomass¹³:

The baseline emissions in case of different fuel in project and baseline scenario is calculated using the below equation:

$$BE_{b,y} = B_{b,y} * ((f_{NRB,y} * EF_{b,fuel,CO2}) + EF_{b,fuel,nonCO2}) * NCV_{b,fuel}$$

Where:

¹³ In case of charcoal fuel consumption is multiplied by 6 as Methodology allows 1 quantity of charcoal is equivalent to 6 quantity of fuel-wood

$BE_{b,y}$	Emissions for baseline scenario b during the year y in tCO ₂ e
$B_{b,y}$	Quantity of fuel consumed in baseline scenario b during year y, in tons
$f_{NRB,y}$	Fraction of biomass used in year y for baseline scenario b that can be established as non-renewable biomass
$NCV_{b,fuel}$	Net calorific value of the fuel that is substituted or reduced (IPCC default for wood fuel, 0.0156 TJ/ton)
$EF_{b,fuel,CO2}$	CO ₂ emission factor of the fuel that is substituted or reduced. 112 tCO ₂ /TJ for Wood/Wood Waste, or the IPCC default value of other relevant fuel
$EF_{b,fuel,nonCO2}$	Non-CO ₂ emission factor of the fuel that is reduced

$$B_{b,y} = N_{p,y} * P_{b,y}$$

Where:

$B_{b,y}$	Quantity of fuel consumed in baseline scenario b during year y, in tons
$P_{b,y}$	Specific fuel consumption for an individual technology in baseline scenario b during year y converted to tons/day
$N_{p,y}$	Project technology-days in the project database for project scenario p through year y

Project Emissions

$$PE_{p,y} = B_{p,y} * ((f_{NRB,y} * EF_{p,fuel,CO2}) + EF_{p,fuel,nonCO2}) * NCV_{p,fuel}$$

$$B_{p,y} = N_{p,y} * ((P_{p,y} * U_{p,y}) + (P_{b,y} * (1 - U_{p,y})))$$

Where:

$PE_{p,y}$	Emissions for project scenario p during the year y in tCO ₂ e
$B_{p,y}$	Quantity of fuel consumed in project scenario p during year y, in tons
$f_{NRB,y}$	Fraction of biomass used in year y that can be established as non-renewable biomass
$NCV_{p,fuel}$	Net calorific value of the project fuel that is substituted or reduced (IPCC default for wood fuel, 0.0156 TJ/ton)

$EF_{p,fuel,CO2}$	CO2 emission factor of the fuel that is substituted or reduced. 112 tCO ₂ /TJ for Wood/Wood Waste, or the IPCC default value of other relevant fuel
$EF_{p,fuel,nonCO2}$	Non-CO2 emission factor of the fuel that is reduced
$N_{p,y}$	Project technology-days in the project database for project scenario p through year y
$P_{p,y}$	Specific fuel consumption for an individual technology in project scenario p during year y converted to tons/day
$U_{p,y}$	Cumulative usage rate for technologies in project scenario j during year y, based on cumulative installation rate and drop-off rate

Leakage

Leakage ($LE_{p,y}$): As per applied GS TPDDTEC methodology version 3.1, leakage emissions are accounted for the following sources:

Leakage emission sources to be assessed:

#	Leakage Source	Assessment
a	The displaced baseline technologies are reused outside the project boundary in place of lower emitting technology or in a manner suggesting more usage than would have occurred in the absence of the project.	No, the displaced baseline technologies (traditional cookstoves) are not reused outside the project boundary. It is highly unlikely that households outside the project boundaries who may have obtained a more efficient stove or other lower emitting technologies (like e.g. LPG gas) would go back to the use of baseline technology.
b	Non-project users who previously used lower emitting energy sources use the non-renewable biomass or fossil fuels saved under the project activity.	No, as it is evident from the baseline survey that the share of household that use a lower emitting energy source such as LPG is very small. These households that use a lower emitting energy source, such as LPG, are not likely to use nonrenewable biomass (NRB) saved under the project activity, instead of lower emitting sources like LPG.
c	The project significantly impacts the NRB fraction within an area where other CDM or VER project activities account for	No, the project leads to reduction in firewood consumption which ultimately brings positive impact to the project

	NRB fraction in their baseline scenario.	vicinity area. It is not expected that the NRB in other areas will be affected so quickly that it would impact other CDM/VER projects activities if any. Furthermore, fNRB update takes place in each renewal of CP which automatically includes the studies evaluating the impact on FNRB.
d	The project population compensates for loss of the space heating effect of inefficient technology by adopting some other form of heating or by retaining some use of inefficient technology.	Not applicable, as the project population location is entire country of Mozambique. Mozambique is an African country crossed by the Equator, where the climate is pleasantly warm, with average temperatures ranging between 20 °C and 25 °C.
e	By virtue of promotion and marketing of a new technology with high efficiency, the project stimulates substitution within households who commonly used a technology with relatively lower emissions, in cases where such a trend is not eligible as an evolving baseline.	Not applicable as the baseline survey conducted showed that end users in the baseline, use traditional cookstoves; hence it is highly unlikely that the project stimulates substitution within households who commonly used a technology with relatively lower emissions.

At the time of monitoring, PP shall be determining if the displaced technologies are used outside the project boundary in place of lower emitting technology through survey. As mentioned in the methodology, Leakage is either calculated as a quantitative emissions volume (tCO₂e) or as a percentage of total emission reductions.

Emission Reductions

The overall GHG reductions achieved by the project activity are then calculated as follows:

$$ER_y = \sum BE_{b,y} - \sum PE_{p,y} - \sum LE_{p,y}$$

Where:

ER _y	Emission reduction for total project activity in year y (tCO ₂ e/yr)
PE _{p,y}	Emissions for project scenario p during the year y in tCO ₂ e
BE _{p,y}	Baseline emissions for baseline scenario b in year y (tCO ₂ e/yr)
LE _{p,y}	Leakage for project scenario p in year y (tCO ₂ e/yr)

The other SDGs impacts of this VPA (SDG 1, SDG 3, SDG 5, SDG 7, SDG 8, SDG 12 and SDG 15) will be estimated as below.

Approach to estimate SDG 1

SDG 1-No Poverty		
HHS _{baseline}	% users reporting money saving due reduced fuel consumption in baseline	0%
HHS _{project}	% users reporting money saving due reduced fuel consumption in in project	100%
Percentage of users reporting money savings due to reduction in purchased fuel consumption in project	(HHS _{project} -HHS _{baseline})	100%

Approach to estimate SDG 3

SDG 3-Good health and well being		
HHsmoke _{baseline}	% users reporting reduction in smoke after shifting to ICS in baseline	0
HHsmoke _{project}	% users reporting reduction in smoke after shifting to ICS in project	100%
Percentage of households reporting reduction in smoke/PM after shifting to ICS in project	(HHsmoke _{project} -HHsmoke _{baseline})	100%

Approach to estimate SDG 5

SDG 5-Gender Equality		
HHtime _{baseline}	% users reporting time saving due to reduced collected fuel consumption / cooking time in baseline	0%
HHtime _{project}	% users reporting time saving due to reduced collected fuel consumption / cooking time in project	95%
Percentage of users reporting time saving due to reduction in collected fuel consumption / cooking time in project	(HHtime _{project} -HHtime _{baseline})	95%

Approach to estimate SDG 7

SDG 7-Affordable and Clean Energy		
HHclean _{baseline}	Number of household having access to affordable and clean energy functional / operational ICS units under Baseline	0
HHclean _{project}	Number of household having access to affordable and clean energy functional / operational ICS units under Project	14,780
Percentage of ICS distributed and operating under the project as an indicator of providing clean technology (relative to baseline stoves).	(HHclean _{project} -HHclean _{baseline})	14,780

Approach related to estimate SDG 8

SDG 8-Decent work and economic growth		
EG _{baseline}	Generation of employment in baseline scenario	0
EG _{project}	Generation of employment in project scenario	40
Total number of jobs created	(EG _{project} -EG _{baseline})	40

Approach related to estimate SDG 12

SDG 12 Responsible consumption and Production		
FC _{baseline}	Quantity of biomass saved per tonnes per year in the baseline	5.11
FC _{project}	Quantity of biomass saved per tonnes per year in the project	2.30
Average percentage of fuel savings reported by users in the project	(1-FC _{baseline} /FC _{project})	55%

Approach related to estimate SDG 15:

SDG 15 Life on land		
FCHH _{baseline}	Average Fuel consumption per HH in baseline (tonnes eq. fuelwood/HH/day)	0.014
FCHH _{project}	Average Fuel consumption per HH in project (tonnes eq. fuelwood/HH/day)	0.0063

Fuelwood equivalent savings reported by users in the project	$(FCHH_{\text{baseline}} - FCHH_{\text{project}}) \times \text{Number of operational ICS}$	113.81
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B.6.2. Data and parameters fixed ex ante

SDG13

Data / parameter ID	ICS 8
Data/parameter	EF_{b,CO_2}
Unit	tCO ₂ /TJ
Description	CO ₂ emission factor arising from use of fuels in baseline Scenario
Source of data	IPCC default value as per 2006 IPCC Guidelines for National Greenhouse Gas Inventories, volume 2, chapter 2 (Table 2.5)
Value(s) applied	Fuelwood (Residential): 112
Choice of data or Measurement methods and procedures	Default IPCC value for fuelwood is applied
Purpose of data	CO ₂ Emission calculation in baseline
Additional comment	-

Data / parameter ID	ICS 9
Data/parameter	$EF_{b,non-CO_2}$
Unit	tCO ₂ /TJ
Description	Non-CO ₂ emission factor arising from use of fuels in baseline scenario
Source of data	As per GS TPDDTEC Meth (AR5 GWP)
Value(s) applied	Fuelwood (Residential): 9.46

Choice of data or Measurement methods and procedures	AR5 GWP
Purpose of data	Non-CO ₂ Emission calculation in baseline
Additional comment	

Data / parameter ID	ICS 10
Data/parameter	EF _{p,CO2}
Unit	tCO ₂ /TJ
Description	CO ₂ emission factor arising from use of fuels in project Scenario
Source of data	IPCC default value as per 2006 IPCC Guidelines for National Greenhouse Gas Inventories, volume 2 (Table 2.5)
Value(s) applied	Fuelwood (Residential): 112
Choice of data or Measurement methods and procedures	Default IPCC value is applied (AR 5)
Purpose of data	CO ₂ Emission calculation in project scenario
Additional comment	-

Data/parameter	EF _{p,non-CO2}
Unit	tCO ₂ /TJ
Description	Non-CO ₂ emission factor for methane arising from use of fuels in project Scenario
Source of data	As per GS TPDDTEC Meth (AR5 GWP)
Value(s) applied	Fuelwood (Residential): 9.46

Choice of data or Measurement methods and procedures	AR5 GWP
Purpose of data	Non-CO ₂ Emission calculation in project scenario
Additional comment	-

Data / parameter ID	ICS 11
Data/parameter	EF _{el,y}
Unit	tCO ₂ /MWh
Description	Emission factor for electricity generation for source j in year y
Source of data	Harmonized Emission Factor data Version 3.1 - UNFCCC ¹⁴
Value(s) applied	0.11078
Choice of data or Measurement methods and procedures	Combined margin CO ₂ emission factor for the project electricity system applicable to all renewable power generation project activities other than wind and solar for the first crediting period
Purpose of data	CO ₂ Emission calculation in project scenario
Additional comment	-

Data/parameter	TDL _{j,y}
Unit	Fraction
Description	Average technical transmission and distribution losses for providing electricity to source j

¹⁴ <https://unfccc.int/climate-action/sectoral-engagement/ifi-harmonization-of-standards-for-ghg-accounting/ifi-twg-list-of-methodologies>

Source of data	CDM Methodological tool 05: Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation, version 03.0
Value(s) applied	0.20
Choice of data or Measurement methods and procedures	Scenario A - Electricity consumption from the grid. The electricity is purchased from the grid only, and either no captive power plant(s) is/are installed at the site of electricity consumption or, if any captive power plant exists on site, it is either not operating or it is not physically able to provide electricity to the electricity consumer;
Purpose of data	CO ₂ Emission calculation in project scenario
Additional comment	-

Data / parameter ID	ICS 12
Data/parameter	NCV _b
Unit	TJ/ton
Description	Net calorific value of the fuels used in the baseline
Source of data	IPCC default 2006, volume 2, chapter 1 (Table 1.2)
Value(s) applied	Fuelwood: 0.0156
Choice of data or Measurement methods and procedures	Default IPCC value for fuelwood is applied
Purpose of data	CO ₂ Emission calculation in project scenario
Additional comment	-

Data / parameter ID	ICS 13
Data/parameter	NCV _p
Unit	TJ/ton

Description	Net calorific value of the fuels used in the project
Source of data	IPCC default 2006, volume 2, chapter 1 (Table 1.2)
Value(s) applied	Fuelwood: 0.0156
Choice of data or Measurement methods and procedures	Default IPCC value for fuelwood, is applied
Purpose of data	CO ₂ Emission calculation in project scenario
Additional comment	-

Data / parameter ID	ICS 17
Data/parameter	$f_{NRB,i,y}$
Unit	Fractional non-renewability
Description	Non-renewability status of woody biomass fuel in scenario i during year y
Source of data	Assessment based on CDM Methodological tool 30: Calculation of the fraction of non-renewable biomass, Version 03.0
Value(s) applied	0.78
Choice of data or Measurement methods and procedures	Assessment based on CDM Methodological tool 30: Calculation of the fraction of non-renewable biomass, Version 03.0
Purpose of data	CO ₂ Emission calculation in project scenario
Additional comment	The f_{NRB} value will remain fixed during the crediting period

Data / parameter ID	ICS 18
Data/parameter	$P_{b,y}$
Unit	tonnes/day

Description	Specific fuel consumption for an individual technology in baseline scenario b during year y converted to tons/day
Source of data	Baseline survey will be conducted at the time of issuance or before and the value will be fixed ex-ante for subsequent issuance conducted and value is fixed ex-ante for subsequent issuance.
Value(s) applied	Fuel wood :0.014
Choice of data or Measurement methods and procedures	Based on baseline Field Test updates. KPT conducted for 3 consecutive dates by measuring fuel and moisture in fuel. Compiled data and averaged the data of 3 days for specific fuel consumption. The details are shown in baseline KPT
	-
Purpose of data	To calculate baseline emissions
Additional comment	Baseline study will be conducted before the first performance review.

SDG 1: No Poverty

Data/parameter	HHS _{baseline}
Unit	Percentage (%)
Description	% users reporting money saving due to reduced fuel consumption in baseline
Source of data	
Value(s) applied	0
Choice of data or Measurement methods and procedures	
Purpose of data	SDG 1 impact calculation
Additional Comment	-

SDG 3: Good Health and Well Being

Data/parameter	HHsmoke _{baseline}
Unit	Percentage (%)
Description	% users reporting reduction in smoke/PM emissions while cooking on improved stove in baseline
Source of data	
Value(s) applied	0
Choice of data or Measurement methods and procedures	
Purpose of data	SDG 3 impact calculation
Additional Comment	

SDG 5: Gender Equality

Data/parameter	HHtime _{baseline}
Unit	Percentage (%)
Description	% users reporting time taken for cooking /fuel collection
Source of data	-
Value(s) applied	0
Choice of data or Measurement methods and procedures	--
Purpose of data	SDG 5 impact calculation
Additional Comment	-

SDG 7: Affordable and clean energy

Data/parameter	HHclean _{baseline}
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Unit	Number
Description	Access to affordable and clean energy (Number of ICS operating)
Source of data	-
Value(s) applied	0
Choice of data or Measurement methods and procedures	
Purpose of data	SDG 7 impact calculation
Additional Comment	-

SDG 8: Decent Work and Economic Growth

Data/parameter	EG _{baseline}
Unit	Number
Description	Quantitative Employment and income generation
Source of data	
Value(s) applied	0
Choice of data or Measurement methods and procedures	-
Purpose of data	SDG 8 impact calculation
Additional Comment	-

SDG 12: Responsible Consumption and Production

Data/parameter	FC _{baseline}
Unit	tonnes/year
Description	Fuel consumption in baseline

Source of data	-
Value(s) applied	5.11
Choice of data or Measurement methods and procedures	--
Purpose of data	SDG 12 impact calculation
Additional Comment	-

SDG 15: Life on Land

Data/parameter	FC _{baseline}
Unit	tonne-eq.fuelwood/HH/day
Description	Average Fuel consumption per HH in baseline
Source of data	-
Value(s) applied	0.014
Choice of data or Measurement methods and procedures	--
Purpose of data	SDG 15 impact calculation

B.6.3. Ex ante estimation of SDG Impact

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Ex-ante calculations of emission reductions related to outcomes of SDG 13 is estimated in ex-ante emission sheet and other SDGs are shown in below table:

SDG targeted	SDG impact indicator	Calculation	Ex-ante estimation
1 No poverty	Indicator: % users reporting money saving due to reduction in purchased fuel	$HHS_{\text{project}} - HHS_{\text{baseline}}$ $= 100\% - 0\%$	100%

	consumption in project		
3 Good health and Well Being	Indicator: % users reporting reduction in smoke/PM after shifting to ICS in project	$\frac{HH_{smoke_{project}} - HH_{smoke_{baseline}}}{HH_{smoke_{baseline}}} = 100\% - 0\%$	100%
5 Gender Equality	Indicator: users reporting time saving due to reduction in collected fuel consumption / cooking time in project	$\frac{HH_{time_{baseline}} - HH_{time_{project}}}{HH_{time_{baseline}}} = 95\% - 0$	95%
7 Affordable and Clean Energy	Indicator: Number of households with operational stoves	$(HH_{clean_{project}} - HH_{clean_{baseline}}) \times \text{Total number of functional / operational ICS distributed}$ $= (100\% - 0\%) \times 14,780$	14,780
8 Decent Work and economic growth	Indicator: Total number of jobs created	$(EG_{project} - EG_{baseline}) = 40 - 0$	40
12 Responsible consumption and production	Indicator: Average % Fuel savings reported by users in the project	$\frac{FC_{project} - FC_{baseline}}{FC_{baseline}} = (1 - 2.30/5.11) \times 100$	55%
15 Life on land	Indicator: Fuelwood equivalent savings reported by user in the project	$(FCHH_{baseline} - FCHH_{project}) \times \text{Total functional device}$ $= (0.014 - 0.0063) \times 14,780$	113.81 tonnes-eq.fuelwood/day
13 Climate Action	Indicator: Total greenhouse gas emissions per year	The overall GHG reductions achieved by the project activity will be calculated as follows:	23,368

		$ER_y = \sum BE_{b,y} - \sum PE_{p,y} - \sum LE_{p,y}$ Where: ER_y Emission reduction for total project activity in year y (tCO₂e/yr) $BE_{b,y}$ Baseline emissions for baseline scenario b in year y (tCO₂e/yr) $PE_{p,y}$ Project emissions for project scenario p in year y (tCO₂e/yr) $LE_{p,y}$ Leakage for project scenario p in year y (tCO₂e/yr) <i>Calculation for ICS:</i> $BE_{b,y} = B_{b,y} * ((fNRB_{b,y} * EF_{b,fuel,CO_2}) + EF_{b,fuel,nonCO_2}) * NCV_{b,fuel}$ Where: $BE_{b,y}$ Emissions for baseline scenario b during the year y in tCO₂e $B_{b,y}$ Quantity of fuel consumed in baseline scenario b during year y, in tons $fNRB$ Fraction of biomass used in year y for baseline scenario b that can be established as	
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		non-renewable biomass $NCV_{b,fuel}$ Net calorific value of the fuel that is substituted or reduced (IPCC default for wood fuel, 0.0156 TJ/ton) $EF_{b,fuel,CO_2}$ CO₂ emission factor of the fuel that is substituted or reduced. 112 tCO₂/TJ for Wood/Wood Waste, or the IPCC default value of other relevant fuel $EF_{b,fuel,nonCO_2}$ Non-CO₂ emission factor of the fuel that is reduced $PE_{p,y} = B_{p,y} * ((fNRB_{p,y} * EF_{p,fuel,CO_2}) + EF_{p,fuel,nonCO_2}) * NCV_{p,fuel}$ Where: $PE_{p,y}$ Emissions for project scenario p during the year y in tCO₂e $B_{p,y}$ Quantity of fuel consumed in project scenario p during year y, in tons $fNRB_{p,y}$ Fraction of biomass used in	
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		year y that can be established as non-renewable biomass $NCV_{p,fuel}$ Net calorific value of the project fuel that is substituted or reduced (IPCC default for wood fuel, 0.0156 TJ/ton) $EF_{p,fuel,CO2}$ CO2 emission factor of the fuel that is substituted or reduced. 112 tCO₂/TJ for Wood/Wood Waste, or the IPCC default value of other relevant fuel $EF_{p,fuel,nonCO2}$ Non-CO2 emission factor of the fuel that is reduced <i>Calculation for EPC:</i> $PE_y = \sum_d EG_{p,d,y} \times EF_{el,y} \times (1 + TDL_{j,y})$ $PE_y = B_{ep,y} * ((f_{NRB,y} * EF_{p,fuel,CO2}) + EF_{p,fuel,nonCO2}) * NCV_{p,fuel}$ Where: $\sum_d$ Sum over all relevant devices PE_y Project emissions from electricity consumption in year y (t CO₂/yr)	
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		$EG_{p,d,y}$ Quantity of electricity consumed by the project device d in year y (MWh/yr) $EF_{el,y}$ Emission factor for electricity system in year y (t CO₂/MWh) $TDL_{j,y}$ Average technical transmission and distribution losses for providing electricity to source j in year y j Sources of electricity consumption in the project PE_y Project emissions from the use of biomass in year y (t CO₂/yr) $B_{ep,y}$ Quantity of fuel consumed by baseline technology in project scenario p during year y, in tons $f_{NRB,y}$ Fraction of biomass used in year y that can be established as non-renewable biomass $NCV_{p,fuel}$ Net calorific value of the fuel that is substituted or	
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		reduced (IPCC default for wood fuel, 0.0156 TJ/ton) $EF_{p,fuel,CO_2}$ CO₂ emission factor of the fuel that is substituted or reduced. 112 tCO₂/TJ for Wood/Wood Waste, or the IPCC default value of other relevant fuel $EF_{p,fuel,nonCO_2}$ Non-CO₂ emission factor of the fuel that is reduced	
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B.6.4. Summary of ex ante estimates of each SDG outcome

SDG 13

Year	Baseline estimate	Project estimate	Net benefit
Year 1	8,490	3,927	4,563
Year 2	38,591	19,488	19,102
Year 2	62,517	31,211	31,306
Year 4	61,745	31,181	30,564
Year 5	62,517	31,211	31,306
Total	2,33,859	1,17,018	1,16,840
Total number of crediting years	5		
Annual average over the crediting period	46,772	23,404	23,368

SDG 1

Year	Baseline estimate	Project estimate	Net benefit
Year 1	0%	100%	100%
Year 2	0%	100%	100%
Year 2	0%	100%	100%
Year 4	0%	100%	100%
Year 5	0%	100%	100%
Total	0 %	100 %	100 %
Total number of crediting years	5		
Annual average over the crediting period	0 %	100 %	100 %

SDG 3

Year	Baseline estimate	Project estimate	Net benefit
Year 1	0%	100%	100%
Year 2	0%	100%	100%
Year 2	0%	100%	100%
Year 4	0%	100%	100%
Year 5	0%	100%	100%
Total	0 %	100 %	100 %
Total number of crediting years	5		
Annual average over the crediting period	0 %	100 %	100 %

SDG 5

Year	Baseline estimate	Project estimate	Net benefit
Year 1	0%	95%	95%
Year 2	0%	95%	95%
Year 2	0%	95%	95%
Year 4	0%	95%	95%

Year 5	0 %	95 %	95 %
Total	0 %	95 %	95 %
Total number of crediting years	5		
Annual average over the crediting period	0 %	95 %	95 %

SDG 7

Year	Baseline estimate	Project estimate	Net benefit
Year 1	0	1,100	1100
Year 2	0	6,100	6100
Year 2	0	14,200	14200
Year 4	0	22,200	22200
Year 5	0	30,300	30300
Total	0	30,300	30300
Total number of crediting years	5		
Annual average over the crediting period	0	14,780	14,780

SDG 8

Year	Baseline estimate	Project estimate	Net benefit
Year 1	0	40	40
Year 2	0	40	40
Year 2	0	40	40
Year 4	0	40	40
Year 5	0	40	40
Total	0	40	40
Total number of crediting years	5		

Annual average over the crediting period	0	40	40
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Total number of jobs generated over 5 years = 70

SDG 12

Year	Baseline estimate	Project estimate	Net benefit
Year 1	5.11	2.30	55%
Year 2	5.11	2.30	55%
Year 2	5.11	2.30	55%
Year 4	5.11	2.30	55%
Year 5	5.11	2.30	55%
Total	5.11	2.30	55 %
Total number of crediting years	5		
Annual average over the crediting period	5.11	2.30	55 %

SDG 15

Year	Baseline estimate	Project estimate	Net benefit
Year 1	0.014	0.006300	8.47
Year 2	0.014	0.006300	46.97
Year 2	0.014	0.006300	109.34
Year 4	0.014	0.006300	170.94
Year 5	0.014	0.006300	233.31
Total	0.014	0.006300	569.03
Total number of crediting years	5		
Annual average over the crediting period	0.014000	0.006300	113.81

B.7. Monitoring plan

B.7.1. Data and parameters to be monitored

SDG 13

Data / parameter ID	ICS 19
Data / Parameter	$P_{p,y}$
Unit	tonnes/day
Description	Specific fuel consumption for an individual technology in project scenario p during year y converted to tons/day
Source of data	Project Kitchen Performance Test
Value(s) applied	ICS – 0.006 tonnes/day EPC – 0.0037 tonnes/day
Measurement methods and procedures	The actual value will be determined based on the Kitchen Performance Test to be carried out in accordance with Annex 4 of the applied methodology TPDDTEC
Monitoring frequency	Every 2 years
QA/QC procedures	Transparent data analysis and reporting
Purpose of data	To calculate emission reductions
Additional comment	A single project fuel consumption parameter is weighted to be representative of the quantity of project technologies of each age being credited in a given project scenario. In case some of the users still continue to use the baseline stoves in spite of community carbon measures to not continue use of the baseline technology, the usage shall be captured during the annual monitoring surveys and will be accounted for in the emission reduction calculations as part of the project Kitchen Performance Test.

Data / parameter ID	ICS 26
Data / Parameter	$U_{p,y}$
Unit	Percentage
Description	Usage rate in project scenario p during year y
Source of data	Annual usage survey
Value(s) applied	90 %
Measurement methods and procedures	Ex-post monitoring and surveys will determine the number of appliances still in operation by field survey

Monitoring frequency	Annual
QA/QC procedures	Transparent data analysis and reporting
Purpose of data	To calculate emission reductions
Additional comment	A single usage parameter is weighted to be representative of the quantity of project technologies of each age being credited in a given project scenario. TPDDTEC Usage Requirements will be followed.

Data / parameter ID	ICS 27
Data / Parameter	$N_{p,y}$
Unit	Project technologies credited (units)
Description	Technologies in the project database for project scenario p through year y
Source of data	Total sales record
Value(s) applied	14,780
Measurement methods and procedures	The total number of appliances deployed during period y is tracked in the Project Database of the specific VPA, which is updated regularly. All appliances distributed will be recorded. Any appliance not recorded in the Project Database will not be credited for emission reductions.
Monitoring frequency	Continuous
QA/QC procedures	Transparent data analysis and reporting
Purpose of data	To calculate emission reductions
Additional comment	The total sales record is divided based on project scenario to create the project database

Data / parameter ID	ICS 13
Data / Parameter	$LE_{p,y}$
Unit	t _{CO₂e} per year
Description	Leakage in project scenario p during year y
Source of data	Baseline and monitoring surveys
Value(s) applied	0

Measurement methods and procedures	Not applicable
Monitoring frequency	Every two years
QA/QC procedures	Transparent data analysis and reporting
Purpose of data	To calculate emission reductions
Additional comment	Aggregate leakage can be assessed for multiple project scenarios, if appropriate

Data / parameter ID	ICS 28
Data / Parameter	$\eta_{new,i}$
Unit	Fraction
Description	Efficiency of the project cookstove
Source of data	Water Boiling Tests (WBT)
Value(s) applied	35%
Measurement methods and procedures	The procedures as outlined in Annex 8 of TPDDTEC will be followed. The minimum sample size of each age group shall comply with the 90/10 rule
Monitoring frequency	Annual
QA/QC procedures	WBT tests will be carried out following the WBT protocol, 4.2.3 (Stove Testing Update - Release of Water Boiling Test Protocol 4.2.3 Clean Cooking Alliance).
Purpose of data	To calculate project fuel consumption
Additional comment	-

Data / parameter ID	ICS 29
Data / Parameter	μ_y
Unit	Fraction
Description	Adjustment to account for any continued use of pre-project devices (baseline stove) in the project scenario during the year y
Source of data	Monitoring/usage surveys
Value(s) applied	0

Measurement methods and procedures	Monitoring surveys to capture cooking habits and stove usage of households in the region, including quantification of use of baseline devices, by formulating questions and/or collecting evidence to determine the frequency of usage of both the project devices and baseline devices
Monitoring frequency	Annual
QA/QC procedures	-
Purpose of data	To calculate emission reduction
Additional comment	-

Data / Parameter	$EG_{P,d,y}$
Unit	MWh/yr
Description	Quantity of electricity consumed by the project electricity consumption source j in year y
Source of data	Calculated
Value(s) applied	0 (PDD ex-ante estimation)
Measurement methods and procedures	Direct measurement method will be used. The project developer will use an appropriate instrument to determine the electricity consumption. The monitoring of quantity of electricity consumed by project activity is determined by conducting 3-day consecutive kitchen performance test in which the energy meter will be fixed to the device to monitor the consumption of electricity for entire day. On the second day, record the final reading before start the cooking. This will be repeated for the third day. The fourth day the final reading will be captured. The average reading of three days provided the consumption of electricity per day. The measurement for three days will then be extrapolated for the entire year. As a cross-check approach, this parameter may be calculated by multiplying wattage of the electric cooking unit with the average number of hours product is used in a year, which will be captured during the monitoring survey. A user will be given a log to record the time of operation in a day for a period of three days similar to project KPT surveys. This will help determine the average

	number of hours ($H_{p,y}$) product is used in a year (pro-rated from three-day study). If the PP chooses the direct measurement method, there is no requirement of measuring $H_{p,y}$ and use of rated wattage.
Monitoring frequency	Annual
QA/QC procedures	-
Purpose of data	To calculate project emissions
Additional comment	The details of calculation are given in ER calculation sheet. The measurements are conservative as it assumes rated wattage of the device based on maximum operational function

Data / Parameter	$H_{p,y}$
Unit	Hours
Description	Project technology-hours in the project database for project scenario p through year y
Source of data	Calculated
Value(s) applied	0 (for PDD ex-ante estimation)
Measurement methods and procedures	The number of hours used will be monitored based on sample basis using simple random sampling. A user will be given a log to record the time of operation in a day for a period of three days. This parameter will be calculated by total number of devices installed in a year with the average number of hours product is used in a year (pro-rated from three-day study), which will be captured during the monitoring survey..
Monitoring frequency	Annual
QA/QC procedures	-
Purpose of data	To calculate project emissions
Additional comment	The measurements are conservative as it assumes rated wattage of the device based on maximum operational function.

SDG 1: End poverty in all its forms everywhere

Data / Parameter	HHS _{project}
Unit	Percentage (%)
Description	% users reporting money savings due to reduction in purchased fuel consumption in project
Source of data	Database
Value(s) applied	100% (for ex-ante estimation)
Measurement methods and procedures	The number of clean energy products distributed are recorded as part of the CME database which will be used to estimate this parameter
Monitoring frequency	Continuous
QA/QC procedures	-
Purpose of data	Monitoring of SDG 1
Additional comment	-

SDG 3: Good Health and Well-being

Data / Parameter	HHsmoke _{project}
Unit	Percentage (%)
Description	% users reporting reduction in smoke/PM after shifting to ICS in project
Source of data	Monitoring Survey
Value(s) applied	100% (for ex-ante estimation)
Measurement methods and procedures	This parameter will be monitored as part of the monitoring survey. Users will be asked as part of monitoring survey if there is less smoke with the use of improved stove.
Monitoring frequency	Once every 2 years (biennially)
QA/QC procedures	-

Purpose of data	Monitoring of SDG 3
Additional comment	The parameter will be measured qualitatively

SDG 5: Gender Equality

Data / Parameter	HHtime _{project}
Unit	Percentage (%)
Description	% users reporting time saving due to reduction in collected fuel consumption/cooking time
Source of data	Monitoring Survey
Value(s) applied	95 % (for ex-ante estimation)
Measurement methods and procedures	This parameter will be monitored as part of the monitoring survey. Users will be asked as part of monitoring survey if they have been able to save time due to use of clean energy products
Monitoring frequency	Once every 2 years (biennially)
QA/QC procedures	-
Purpose of data	Monitoring of SDG 5
Additional comment	The parameter will be measured qualitatively

SDG 7: Affordable and Clean Energy

Data / Parameter	HHclean _{project}
Unit	Number
Description	Number of households with operational stoves
Source of data	CME Database
Value(s) applied	14,780
Measurement methods and procedures	The number of improved cookstoves distributed will be recorded as part of the CME database which will be used to estimate this parameter
Monitoring frequency	Continuous

QA/QC procedures	-
Purpose of data	Monitoring of SDG 7
Additional comment	-

SDG 8: Decent Work and Economic Growth

Data / Parameter	EG _{project}
Unit	Number
Description	Total no of jobs created
Source of data	CME Database
Value(s) applied	40 (for ex-ante)
Measurement methods and procedures	CME will maintain a record of jobs created as a result of implementation of this programme in form employee list/ payroll system/ contracts/ pay slips
Monitoring frequency	Continuous
QA/QC procedures	-
Purpose of data	Monitoring of SDG 8
Additional comment	-

SDG 12: Responsible Consumption and Production

Data / Parameter	FC _{project}
Unit	tonnes/capita/year
Description	Consumption of non-renewable biomass per household
Source of data	Monitoring survey
Value(s) applied	2.30
Measurement methods and procedures	This will be recorded during KPT conducted in HH.
Monitoring frequency	Annually
QA/QC procedures	-
Purpose of data	Monitoring of SDG 12

Additional comment	-
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SDG 15: Life on Land

Data / Parameter	FCHH _{project}
Unit	Tonnes eq. fuelwood/day
Description	Average Fuel consumption per HH in project scenario
Source of data	-
Value(s) applied	0.0063
Measurement methods and procedures	-
Monitoring frequency	Annually
QA/QC procedures	-
Purpose of data	Monitoring of SDG 12 and 15
Additional comment	-

B.7.2. Sampling plan

>>

The sampling plan given below defines the approach for determining relevant monitoring parameters via ex-post sampling. All required monitoring and documentation will be implemented, reported, consolidated and managed by the CME or a qualified expert partner to meet verification requirements.

Target population– The target population is the total stove population (residential ICS and EPC) existing under the VPA boundary. If multiple VPAs are grouped for cross-VPA sampling, then the target population would encompass the total ICS population served under the grouped VPAs, as recorded in the Distribution Database/Record.

Sampling Plan across VPAs within the PoA

There are two project scenarios under the VPA – domestic improved stove and electric pressure cooker.

Improved Cookstove

Project technologies especially the domestic stoves which are part of the VPA have similar design and performance characteristics. All have same fundamental combustion technology and their respective thermal efficiencies or specific consumptions do not differ by more than +/-5% in absolute terms which can be verified by the technical specification sheet. Hence, these will be included under a single project scenario.

Electric Pressure Cooker

EPC will be considered as a separate project scenario. Sampling will be undertaken separately for different project scenarios.

If homogeneity of population can be demonstrated, or differences are taken into account (stratification) in the sample size calculation then 90/10 confidence/precision will be applied. In case of cross-VPA sampling, 95/10 confidence/precision will be applied.

Monitoring / Usage survey:

The monitoring survey investigates changes over time in a project scenario by surveying end users with project technologies on an annual basis. An usage survey determines the drop off rates as project technologies age and users switch back to the baseline technology.

Random sampling shall be applied for determining the usage rate of for each project scenario separately. The sample size determined shall be distributed within each age category based on percentage of stoves in corresponding age category. For example, if only technologies in the first year of use (age 0-1) are being credited, a usage parameter must be established through a usage survey for technologies age 0-1. If technologies in the first year of use (age 0-1) and second year of use (age 1-2) are credited, a usage parameter is required that is weighted to be representative of drop off rates for technologies age 0-1 and age 1-2. The minimum total sample size is 100, with at least 30 samples for project technologies of each age being credited.

The majority of interviews in a monitoring and usage survey will be conducted by community carbon staff or by hired experts who would be trained before and include expert observation by the interviewer within the kitchen in question, while the remainder may be conducted via telephone by the same interviewers on condition that

in-kitchen observational interviews are first concluded and analyzed such that typical circumstances are well understood by the telephone interviewers.

Project and Baseline KPT:

For determining the fuel consumption in the baseline / project scenarios, the random sampling for KPTs shall be applied. The sample size determined shall be distributed within each age category based on percentage of stoves in corresponding age category. For determining the fuel consumption in the baseline / project scenario the KPT sample size determination shall be based on Kitchen Performance Test Protocol. Alternatively, the guideline given in TPDDTEC version 3.1 may be used for determination of sample size.

The project fuel consumption test can also be replaced with ageing test approach from second monitoring period.

In case PP opts for the ageing test approach instead of biennial project KPTs, annual WBTs would be conducted on a representative sample of each age group. The sample size would be calculated using student's t-distribution approach for mean based parameter. The sample size are appropriately sampled from each age group and enough so that the results comply with the 90/10 rule. In case of not meeting the required confidence/precision, lower bound value will be used. The WBT shall be carried out along with the project KPTs prior to 1st issuance and then subsequently WBTs shall be carried out annually to monitor the degradation in the efficiency of the ICS. The WBTs should be conducted in the last 3 months of the monitoring period or after the monitoring period, provided it is representative of each age group.

Sampling Plan for monitoring parameters

The objective of the sampling plan for this VPA is to determine:

- (i) $\eta_{new,i}$: Efficiency of the project cookstove. This parameter will be monitored with 90% confidence and 10% precision annually for each VPA, or 95/10 in case of cross-VPA sampling or biennial monitoring (when applicable).
- (ii) $U_{p,y}$: Usage rate in project scenario p during year y. This would be based on usage survey, to be conducted upon the representative sample of operational stoves.
- (iii) μ_y : Adjustment to account for any continued use of pre-project devices (baseline stove) in the project scenario during the year y. Survey

questionnaires administered to a sample of end users will elicit visual inspections of the household and if necessary an interview to confirm whether they are still using a baseline stove and, in that case, to obtain self-reported estimates of the amount of non-renewable biomass consumed per day in traditional stoves in parallel to the project stove.

- (iv) $P_{p,y}$: Specific fuel consumption for an individual technology in project scenario. The actual value will be determined based on the Kitchen Performance Test to be carried out in accordance with Annex 4 of the applied methodology TPDDTEC version 3.1.
- (v) $EG_{P,d,y}$: The monitoring of quantity of electricity consumed by project activity will be done for 3-consecutive days similar to kitchen performance test in which the energy meter will be fixed to the device to monitor the consumption of electricity for entire day. On the fourth day the final reading will be captured.
- (vi) $H_{p,y}$: Project technology-hours for project database for project technology. This parameter will be monitored with 90% confidence and 10% precision annually for each VPA, or 95/10 in case of cross-VPA sampling or biennial monitoring (when applicable)¹⁵.

B.7.3. Other elements of monitoring plan

>>

Monitored Systems

1. Total Distribution Record: The total distribution record documents the information listed below for the technologies implemented. A carbon waiver including a warranty card will be distributed with each stove distributed. The CME makes every effort to retrieve this information (paper form or electronically (i.e. SMS) but cannot guarantee the collection of information for waivers and warranties with every stove due to challenges such as high rates of illiteracy and logistical challenges. The total distributions record will be kept electronically and with supporting evidence from

¹⁵ As this is a large-scale VPA, the CME will not follow cross-VPA sampling for monitoring.

paper records and/or SMS tracking records and will be provided to the GS-VVB at verification. The Total Distribution Record contains:

- a) VPA-ID (VPA to which appliance belongs)
- b) Unique identification of stove using stove serial number
- c) Partner organization name, address and telephone (as available)
- d) Date of distribution and model/type of project technology distributed
- e) Quantity of project technology distributed as evidenced by invoices

Frequency: Ongoing

2. Project Database: [Parameter $N_{p,y}$] Each VPA will have a specific Project Database that records each ICS crediting in that VPA. Every ICS listed in the Total Distribution Record will be transferred into the Project Database of this VPA as needed to expand the number of ICS deployed, until the maximum threshold for this VPA is reached. In addition to the information provided in the Total Distribution Record, the VPA-specific Project Database will record user details (enough for end-user identification and follow-up) for all, or a subset of all, appliances deployed. End-user details recorded are:

- a) Name
- b) Telephone, or address (as available)
- c) Type of stove (stove model) and fuel the ICS is replacing: Example – traditional or improved baseline stoves, or wood or charcoal fuel.

ICS with end-user details recorded here will be used to determine other information needed using as many samples as commensurate with representative sampling to calculate the emission reductions of the project, following TPDDTEC v3.1-page 28/29 guidance on Total Sales Record.

Frequency: Ongoing

3. Continued use of displaced traditional stoves

The replaced low efficiency devices are disposed of and not used within the boundary or within the region. Monitoring surveys conducted on households (HH) using ICS will also investigate the extent to which baseline traditional stoves are

still in use. If it is found that a traditional stove is still used, even in a secondary role, the HH will be encouraged to discard their traditional stove through the Disposal Policy. Evidence of use of traditional stoves in other kitchens is not possible as the stoves in question are fixed type. As a final resort, in cases where further implementation of the Disposal Policy is shown by the monitoring visits to be ineffective, emission reductions are calculated taking into account only that portion of the wood used in the kitchen overall is subject to the improved efficiency of the ICS.

4. Disposal Policy: The project activity encourages full adoption of project stove distributed. To incentivize full adoption, the project proponents extend a full-year warranty in exchange for the disposal of a household's traditional stove.

5. Organizational structure of monitoring and inclusions:

Person	Role
CME database administrator	The database administrator is responsible for updating and maintaining all electronic databases. Required competencies include experience with data management systems (eg. Excel, STATA, or SPSS), minimum 2 years working experience in a similar field, and at minimum a bachelor's degree from an institution of higher education.
Monitoring team	The monitoring team will be assigned by the CME to conduct the user interviews and appliance tests during the periodic sampling and reports the results to the database administrator. The skills and experience required for the data collection activities include: • Experience conducting surveys/tests • Experience conducting door-to-door surveys of biomass consumption • Local language skills (especially important for input to questionnaire design and interviewing of end users) • English language skills

	• Cultural awareness • Numerical proficiency • Data entry skills
--	--

6. Quality Assurance / Quality control

As the PoA is intended to include multiple regions within a country with a high level of cultural diversity as well as different end user groups, there is no “one size fits all” approach for dealing with these issues. However, in order to avoid many of these problems the CME will undertake the following strategies, tailoring the specific approach to the local circumstances:

- a. Ensuring end user awareness. At the time of distribution, the ICS customer is made aware that they are required to participate in monitoring activities. This will be via a written statement (in English and local language where appropriate) on the carbon waiver form, or via alternative means such as training distribution personnel to explain the importance of monitoring to each customer.
- b. Questionnaire design. The design of the questionnaire will ensure that the questions are non-intrusive and easy to understand for both the interviewee and interviewer. For example, when conducting sampling to estimate the parameter μ_y a simplified approach has been designed to avoid the need for asking customers how much money they spend on fuel.
- c. Drawing on local knowledge. The local contractors to be hired by the CME in the country will play an important role in tailoring the approach to suit local circumstances. For example, in some instances, it may be essential for a local person to conduct the interview in order to obtain accurate results, e.g. to explain to the end user that their old stove will not be removed if they admit to its continued use.
- d. Quality of contractors. Any third parties hired by the CME to carry out sampling will be required to demonstrate a high level of cultural

awareness, local language skills and appropriate experience with data entry and data management. The CME will ensure that contractors are adequately trained for the tasks they are contracted for (eg. carrying out of WBTs in line with a methodology supported by an appropriate international body/standards). Training will also be provided on how to deal with non-responses, refusals and other problems should these occur.

7. Fuel switch scenario

The potential increase in fuels with different carbon intensities (such as increased charcoal usage on users that consumed originally firewood prior to the purchase of the ICS) will be accounted by monitoring consumption of biomass distinguishing the different fuels as charcoal and firewood. In light of this the future monitoring tasks are to be performed accordingly.

SECTION C. DURATION AND CREDITING PERIOD

C.1. Duration of project

C.1.1. Start date of VPA

>>

01/01/2022

C.1.2. Expected operational lifetime of VPA

>>

15 years (3 x 5 years)

C.2. Crediting period of project

C.2.1. Start date of crediting period

>>

05/07/2022

C.2.2. Total length of crediting period

>>

5 years. The crediting period may be renewed twice in line with the Community Services Activity Requirements.

SECTION D. SUMMARY OF SAFEGUARDING PRINCIPLES AND GENDER SENSITIVE ASSESSMENT

D.1. Safeguarding Principles that will be monitored

A completed Safeguarding Principles Assessment is in [Appendix 1](#), no ongoing monitoring is applicable to this VPA.

D.2. Assessment that project complies with GS4GG Gender Sensitive requirements

Question 1 - Explain how the project reflects the key issues and requirements of Gender Sensitive design and implementation as outlined in the Gender Policy?	The local stakeholder consultation meeting will be carried out following a gender sensitive approach. The VPA refers to raising awareness and consideration of gender norms, roles and relations but do not necessary address inequality generated by unequal norms, roles, and relations through remedial action beyond creating gender awareness. The project also includes compliance with the gender 'do no harm' safeguard, gender-gap analysis and gender sensitive stakeholder consultation.
Question 2 - Explain how the project aligns with existing country policies, strategies and best practices	The Climate Change and Gender Action Plan (ccGAP) aims to comprehensively develop and integrate the gender perspective throughout the environmental sector to improve the quality of life for all Mozambicans, and in particular for women and local communities. Mozambique ranks 174 out of 188 countries for per capita GHG 16emissions and contributes 0.06% to global

¹⁶ https://reliefweb.int/sites/reliefweb.int/files/resources/Mozambique_4.pdf

emissions¹⁷. However, for vulnerability to climate change, Mozambique ranks 160 out of 181 countries in the ND-GAIN index³ (2016). It is the 35th most vulnerable and the 24th least ready country – meaning that it is vulnerable to, yet unready to address climate change effects.

In 2010, the Republic of Mozambique, thanks to the leadership of the Ministry for Coordination of Environment Affairs (MICOA), became the first country in the world to pursue a national strategy and associated action plan for ensuring that its environment and climate change policies and programs are gender responsive.

The strategy's mission has been to promote gender equality and equity and to improve participation of women and the poorest communities in the preservation of natural resources, environmental management and climate change mitigation and adaptation action, through their empowerment

As established in the National Climate Change Adaptation and Mitigation Strategy (NCCAMS) (MICOA, 2012), the national priority is defined in its mission “to increase resilience in the communities and the national economy including the reduction of climate risks and promote a low carbon development and the green economy through the integration of adaptation and mitigation in sectorial and local planning”.

The implementation of the actions referred will limit the GHG emissions by sources and the removals by sinks at the same time as they contribute to the increase of the well-being of the Mozambicans through the increase of the

¹⁷ https://reliefweb.int/sites/reliefweb.int/files/resources/Mozambique_4.pdf

	access to renewable energy sources and willing to participate in the market mechanisms to be established which would allow access to clean technologies in order to mitigate the emissions arising from exploiting, managing and using the natural capital that is available. Mozambique aims, on a preliminary basis, a total reduction of about 76,5 MtCO₂ -eq in the period from 2020 to 2030 compared to a Business-as-Usual (BAU) scenario with 23.0 MtCO₂ -eq by 2024 and 53.4 MtCO₂ -eq from 2025 to 2030. The Mozambique INDC highlights that the implementation of any proposed reduction is conditional on the provision of financial, technological, and capacity-building support from the international community. The country strategy expanded the focus from primary adaptation to mitigation, capacity building, and financing. Its three objectives are: • Adaptation: to become resilient to impacts of climate change, while minimizing climate risks to people and property. • Mitigation: to identify and implement opportunities to reduce GHG emissions. Capacity and resources: To build institutional and human capacity and explore opportunities to access technology and financial resources to implement this strategy. The PoA would bring access to international finance for successful implementation of Mozambique conditional NDC target.
Question 3 - Is an Expert required for the Gender Safeguarding Principles & Requirements?	Not required. Improved cookstove projects not following Gender responsive approach do not require to contract an expert as per Gender Equality Requirements & Guidelines. The project will have a positive impact on women included under the programme by providing them with cleaner cooking solutions.

Question 4 - Is an Expert required to assist with Gender issues at the Stakeholder Consultation?	Not required. Improved cookstove projects not following Gender responsive approach do not require to contract an expert as per Gender Equality Requirements & Guidelines. The stakeholder consultation will include interactions with potential beneficiaries including women and their feedback shall be considered appropriately.
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SECTION E. SUMMARY OF LOCAL STAKEHOLDER CONSULTATION

The below is a summary of the 2 step GS4GG Consultation for monitoring purposes. Please refer to the separate Stakeholder Consultation Report for a complete report on the initial consultation and stakeholder feedback round.

E.1. Summary of stakeholder mitigation measures

>>

The physical local stakeholder consultation meeting was conducted on 28/02/2022. Invitations were sent to stakeholders along with feedback form and non-technical summary via email on 28/01/2022. Also, newspaper advertisement was published on 29/01/2022. The screenshots of the advertisement are shown below. Apart from e-mail and newspaper advertisement, stakeholders were invited via informal channels such as phone, whatsapp. The feedback forms were distributed during the physical meeting as well as via email along with invitation

Sub: Invitation to Initial Stakeholder Consultation for UpEnergy – Social and Climate Impact Programme

YAMM - Sub: Invitation to Initial Stakeholder Consultation for UpEnergy – Social and Climate Impact Programme ✕

UpEnergy Carbon Technical Team <technical@upenergygroup.com>
to Bruno Merven ▾

9:47 PM (39 minutes ago) ☆ ↶ ⋮

Dear Bruno Merven,

Greetings from The UpEnergy Group, Mozambique.

UpEnergy Group is the Co-ordinating/Managing entity of GS ID 10963, UpEnergy – Social and Climate Impact Programme

The PoA (and its VPAs 1 to 14) is registered under Global Standard for the Global Goals (GS4GG) to certify the contribution of PoA towards achievement of various SDGs. The purpose of the Gold Standard Programme of Activities (PoA) is to provide end-users with energy-efficient cooking technologies and Safe water systems (WPS) with the aim to reduce greenhouse gas (GHG) emissions from the burning of non-renewable woody biomass and/or charcoal for cooking and water treatment to make it safe for consumption. Thus, the PoA supports the intended goals of reducing fuel consumption, improving health, reducing deforestation and reducing GHG emissions in Mozambique

As a valued stakeholder of the PoA, you are requested to participate in the design of an initiative intended to create, register and trade carbon credits in order to fund the expanded dissemination of improved cooking stoves and water purification systems. The stakeholder consultation is an opportunity at the early stage to present the projects to the relevant parties involved and include their feedback into the design of the projects.

We are currently beginning the initial stakeholder consultation process. You are requested to attend the physical meeting which will be held on **28th February 2022 at Pensao Martins, Av 24 de Julho, Maputo**. We welcome your feedback and suggestions and cordially request you to fill the form to share your views on this PoA.



The SFR was initiated on 28/02/2022. The reminder email for SFR closure was sent on 08/05/2022. The SFR closing mail was sent on 22/05/2022. The screenshot of the emails are below:

Closure of Stakeholder Feedback Round (Mozambique)- GS ID 10963

YAMM - Closure of Stakeholder Feedback Round (Mozambique)- GS ID 10963 x



UpEnergy Carbon Technical Team <technical@upenergygroup.com>

12:14 PM (3 minutes ago)



to Lloyd

Dear Lloyd Archer

Thanks a lot for your time and feedback.

With respect to the subject matter, please note that the Stakeholder Feedback Round (SFR) is now officially closed as of 23rd May 2022. Many thanks for your time and valuable feedback.

With Regards

UpEnergy Group

Mozambique

The consultation involved inviting comments/feedback from the following, but limited to, category of stakeholders:

- Local people impacted by the project and official representatives.
- Local policy makers and representatives of local authorities
- Local non-governmental organizations working on topics relevant to the PoA
- Host country DNA
- Research organizations/institutes working on the technology involved in the PoA
- Renewable energy/energy development nodal agencies
- GS NGO Supporters (present in that host country) + International GS NGO Supporter NGOs
- Other relevant local NGOs of the host country

The consultation process included invitation to wide range of invitees to include effective and equal participation of both men and women. But due to COVID, the attendees were less. Prior to consultation, Community Carbon had provided with the following documents in the language that allows local stakeholders to understand and engage with the project.

- Non-Technical summary with relevant information
- Summary of the economic, social and environmental impacts of the project
- Contact details of the Community Carbon to get for getting technical and project related information of the project
- Email id of Community Carbon was provided to the stakeholders who are not able to join the consultation meeting

The stakeholder consultation report along with findings have been submitted separately. Summary of the few findings from LSC and SFR are listed below:

Organization	Stakeholder Comments (Relevant)	If any changes in project design were made
African Clean Energy	Ensure that the data collection on SDGs are done correctly and people are really benefitting from the programme	NA
African Clean Energy	Programme will ensure to bridge the gap of gender equality and provide opportunities to women	NA
End User	The programme may lead to monetary saving	NA
Bold Energy Limited	Programme provides access to affordable clean energy products to poor households and overall has positive impact on environment	NA

E.2. Final continuous input / grievance mechanism

Method	Include all details of Chosen Method (s) so that they may be understood and, where relevant, used by readers.
Continuous Input / Grievance Expression Process Book (mandatory)	In line with section 2.1 of the Annex W Expression book has been placed at office of community carbon in Mozambique. Stakeholders are free to voice their concerns via the Grievance Expression Book. By maintaining feedback book at the local office, it is ensured that stakeholders that don't have access to electronic media for expressing concerns / grievances are also able to share their concerns / feedback. Additionally, the end users always have an option to revert to the salesperson (representative of distribution/retail partners etc.) in case of any feedback / complaints with the product post distribution.

GS Contact
(mandatory) help@goldstandard.org

Other Email: anantha@communityco2.

SECTION F. **Eligibility and inclusion criteria for VPAs inclusion**

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Please Refer Section A.1.1

APPENDIX 1 - SAFEGUARDING PRINCIPLES ASSESSMENT

Complete the Assessment below and copy all Mitigation Measures for each Principle into [SECTION D](#) above.

Assessment Questions / Requirements	Justification of Relevance (Yes / potentially / no)	How Project will achieve Requirements through design, management or risk mitigation.	Mitigation Measures added to the Monitoring Plan (if required)
Principle 1. Human Rights			
1. The Project Developer and the Project shall respect internationally proclaimed human rights and shall not be complicit in violence or human rights abuses of any kind as defined in the Universal Declaration of Human Rights 2. The Project shall not discriminate with regards to participation and inclusion	No	The project will be implemented in collaboration with local partners and CME will respect internationally proclaimed human rights and shall not be complicit in violence or human rights abuses of any kind as defined in the Universal Declaration of Human Right. The project will not discriminate with regards to participation and inclusion.	N/A
Principle 2. Gender Equality			
1. The Project shall not directly or indirectly lead to/contribute to adverse	No.	The project activity doesn't endorse any form of discrimination based on	N/A

impacts on gender equality and/or the situation of women - Projects shall apply the principles of non-discrimination, equal treatment, and equal pay for equal work - The Project shall refer to the country's national gender strategy or equivalent national commitment to aid in assessing gender risks (where required) Summary of opinions and recommendations of an Expert Stakeholder(s)		gender. ICS will be distributed to all willing customers within the project boundary. The project will have a positive impact on women considering that they will spend less time on cooking or fuel procurement and will be able to cook in cleaner environment. In the revised Mozambican Constitution established in 2004, several clauses emphasize gender equality. Article 35 stresses on universality without discrimination (including sex), and Article 36 promises gender equality, while Article 122 stipulates promotion and support for women's participation, role, and empowerment in all spheres of the country's political,	
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		economic, social and cultural life ¹⁸	
Principle 3. Community Health, Safety and Working Conditions			
1. The Project shall avoid community exposure to increased health risks and shall not adversely affect the health of the workers and the community	No	The project doesn't expose the community to increased health risks and is not adversely affecting the health of workers and the community. Use of ICS will contribute in improving the health of users as compared to inefficient traditional stoves by reducing the indoor air pollution. The improved cookstoves are designed, engineered and manufactured in Uganda is being distributed under the PoA. UpEnergy in its capacity as coordinating / managing entity of the PoA confirms that	N/A

¹⁸ https://www.jica.go.jp/english/our_work/thematic_issues/gender/background/c8h0vm0000anjqj6-att/mozambique_2015.pdf

		the workers participating in the project activity are not exposed to unsafe or unhealthy work environments and applicable country regulations are strictly adhered to. The same has been vetted during selection of manufacturing partner by the CME. A declaration to this effect signed by CME has been submitted to Gold Standard.	
Principle 4.1 Sites of Cultural and Historical Heritage			
Does the Project Area include sites, structures, or objects with historical, cultural, artistic, traditional or religious values or intangible forms of culture?	No	Since this is a cookstove project, there is no risk of risk to cultural and historic heritage.	N/A
>>			
Principle 4.2 Forced Eviction and Displacement			
Does the Project require or cause the physical or economic relocation of peoples (temporary or permanent, full or partial)?	No	Since this is a cookstove project, there is no risk of forced eviction and displacement.	N/A
>>			

Principle 4.3 Land Tenure and Other Rights			
a. Does the Project require any change, or have any uncertainties related to land tenure arrangements and/or access rights, usage rights or land ownership?	No	Since this is a cookstove project, there is no risk of uncertainty due to land rights/ownership.	Not required
b. For Projects involving land use tenure, are there any uncertainties with regards to land tenure, access rights, usage rights or land ownership?	No	Not Relevant	Not required
> >			
Principle 4.4 Indigenous People			
a. Are indigenous peoples present in or within the area of influence of the Project and/or is the Project located on land/territory claimed by indigenous peoples?	No	The project does not affect indigenous population directly or indirectly	Not Required
> >	No		
Principle 5. Corruption			

1. The Project shall not involve, be complicit in or inadvertently contribute to or reinforce corruption or corrupt Projects	No	CME will ensure that the project doesn't involve, be complicit in or inadvertently contribute to or reinforce corruption or corrupt Projects.	N/A
Principle 6.1 Labour Rights			
1. The Project Developer shall ensure that all employment is in compliance with national labour occupational health and safety laws and with the principles and standards embodied in the ILO fundamental conventions 2. Workers shall be able to establish and join labour organisations 3. Working agreements with all individual workers shall be documented and implemented and include: a) Working hours (must not exceed 48 hours	No	1. The project is implemented on the ground by the UPE in collaboration with other project partners. The project employment will be in compliance with national labour occupational health and safety laws and with the principles and standards embodied in the ILO fundamental conventions. 2. The workers employed by CME for the project are able to establish and join labour organizations. 3. The working agreements with the individual workers will be documented and implemented and the minimum requirements	

per week on a regular basis), AND b) Duties and tasks, AND c) Remuneration (must include provision for payment of overtime), AND d) Modalities on health insurance, AND e) Modalities on termination of the contract with provision for voluntary resignation by employee, AND f) Provision for annual leave of not less than 10 days per year, not including sick and casual leave. 4. No child labour is allowed (Exceptions for children working on their families' property requires an <u>Expert Stakeholder</u> opinion) 5. The Project Developer shall ensure the use of		stated will be respected as applicable. 4. The minimum age for possible staff hired is 18. 5. All the workers will be provided with appropriate equipment, training documentation and reporting of accidents and incidents, and emergency preparedness and response measures.	
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appropriate equipment, training of workers, documentation and reporting of accidents and incidents, and emergency preparedness and response measures			
Principle 6.2 Negative Economic Consequences			
1. Does the project cause negative economic consequences during and after project implementation?	No	The project involves sale of ICS to willing customers within the project boundary. Carbon revenues are important for creating awareness among the end users and strengthening the local sales and distribution services.	N/A
>>			
Principle 7.1 Emissions			
Will the Project increase greenhouse gas emissions over the Baseline Scenario?	No	The project will reduce the GHG emissions which will be monitored and verified in line with the GS4GG.	N/A
>>			
Principle 7.2 Energy Supply			
Will the Project use energy from a local grid or power supply (i.e., not connected to a	No	The project does not use energy from a local grid or power supply. Use	N/A

national or regional grid) or fuel resource (such as wood, biomass) that provides for other local users?		fuelwood/ charcoal will be significantly reduced by introducing Improved Cookstoves.	
>>			
Principle 8.1 Impact on Natural Water Patterns/Flows			
Will the Project affect the natural or pre-existing pattern of watercourses, ground-water and/or the watershed(s) such as high seasonal flow variability, flooding potential, lack of aquatic connectivity or water scarcity?	No	The project is a cookstoves distribution programme and will not affect the natural or pre-existing pattern of watercourses, ground-water and/or the watershed(s) such as high seasonal flow variability, flooding potential, lack of aquatic connectivity or water scarcity.	N/A
>>			
Principle 8.2 Erosion and/or Water Body Instability			
Could the Project directly or indirectly cause additional erosion and/or water body instability or disrupt the natural pattern of erosion?	No	The project is a cookstoves distribution programme and will not directly or indirectly cause additional erosion and/or water body instability or disrupt the natural pattern of erosion.	N/A
>>			
Principle 9.1 Landscape Modification and Soil			

Does the Project involve the use of land and soil for production of crops or other products?	No	The project is a cookstoves distribution programme and does not involve the use of land and soil for production of crops or other products.	N/A
>>			
Principle 9.2 Vulnerability to Natural Disaster			
Will the Project be susceptible to or lead to increased vulnerability to wind, earthquakes, subsidence, landslides, erosion, flooding, drought or other extreme climatic conditions?	No	The project is a cookstoves distribution programme and will not be susceptible to or lead to increased vulnerability to wind, earthquakes, subsidence, landslides, erosion, flooding, drought or other extreme climatic conditions.	N/A
>>			
Principle 9.3 Genetic Resources			
Could the Project be negatively impacted by or involve genetically modified organisms or GMOs (e.g., contamination, collection and/or harvesting, commercial development, or take place in facilities or farms that include GMOs in their processes and production)?	No	The Project is not negatively impacted by the use of genetically modified organisms or GMOs.	N/A

>>			
Principle 9.4 Release of pollutants			
Could the Project potentially result in the release of pollutants to the environment?	No	As the project includes distribution of improved stoves, there is no risk of releasing increased pollutants to the environment.	N/A
>>			
Principle 9.5 Hazardous and Non-hazardous Waste			
Will the Project involve the manufacture, trade, release, and/ or use of hazardous and non-hazardous chemicals and/or materials?	No	The Project does not involve the manufacture, trade, release, and/or use of hazardous chemicals and or materials.	N/A
>>			
Principle 9.6 Pesticides & Fertilisers			
Will the Project involve the application of pesticides and/or fertilisers?	No	The project does not involve the application of pesticides and/or fertilisers.	N/A
>>			
Principle 9.7 Harvesting of Forests			
Will the Project involve the harvesting of forests?	No	The project does not involve involve the harvesting of forests.	N/A
>>			
Principle 9.8 Food			

Does the Project modify the quantity or nutritional quality of food available such as through crop regime alteration or export or economic incentives?	No	The project does not modify the quantity or nutritional quality of food available such as through crop regime alteration or export or economic incentives.	N/A
>>			
Principle 9.9 Animal husbandry			
Will the Project involve animal husbandry?	No	The project does not involve animal husbandry.	N/A
>>			
Principle 9.10 High Conservation Value Areas and Critical Habitats			
Does the Project physically affect or alter largely intact or High Conservation Value (HCV) ecosystems, critical habitats, landscapes, key biodiversity areas or sites identified?	No	The project is a cookstoves distribution programme and physically affect or alter largely intact or High Conservation Value (HCV) ecosystems, critical habitats, landscapes, key biodiversity areas or sites identified.	N/A
>>			
Principle 9.11 Endangered Species			
Are there any endangered species identified as potentially being present within the Project boundary (including those that may route through the area)?	No	The project boundary is geographical sites of cookstoves distributed and there are no endangered species identified as	N/A

AND/OR Does the Project potentially impact other areas where endangered species may be present through transboundary affects?		potentially being present within the Project boundary.	
> >			

APPENDIX 2- CONTACT INFORMATION OF VPA IMPLEMENTER

Organization name	Community Carbon
Registration number with relevant authority	188181
Street/P.O. Box	
Building	
City	PORT LOUIS
State/Region	
Postcode	
Country	MAURITIUS
Telephone	
E-mail	anantha@communityco2.org
Website	
Contact person	
Title	DIRECTOR
Salutation	
Last name	RAGNAR BURNING
Middle name	
First name	KRISTIAN
Department	
Mobile	
Direct tel.	
Personal e-mail	

APPENDIX 3- LUF ADDITIONAL INFORMATION

Risk of change to the Project Area during Project Certification Period:	NA
Risk of change to the Project activities during Project Certification Period:	NA
Land-use history and current status of Project Area:	NA
Socio-Economic history:	NA
Forest management applied (past and future)	NA
Forest characteristics (including main tree species planted)	NA
Main social impacts (risks and benefits)	NA
Main environmental impacts (risks and benefits)	NA
Financial structure	NA
Infrastructure (roads/houses etc):	NA
Water bodies:	NA
Sites with special significance for indigenous people and local communities - resulting from the Stakeholder Consultation:	NA
Where indigenous people and local communities are situated:	NA
Where indigenous people and local communities have legal rights, customary rights or sites with special cultural, ecological, economic, religious or spiritual significance:	NA

APPENDIX 4-SUMMARY OF APPROVED DESIGN CHANGES

Please refer to [Design Changes Requirements](#) for more information on procedures governing Design Changes